

LEARNER GUIDE

Metabo-Diet

*Harmonizing Dietary and Exercise Phenotypes
with Metabolomics Across CFDE Resources*

A guided, self-paced module

Release 1.0 | August 2026

Prepared for CFDE learners working with public metabolomics data

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How to use this guide

The module takes approximately 153 minutes after software setup: 140 minutes of lesson activity, a 5-minute pretest, and an 8-minute posttest. First-time Python or metabolomics learners should allow another 30 to 60 minutes. Extract `metabo_diet_analysis_bundle.zip` and `metabo_diet_templates.zip`, and keep this guide plus the three worksheets open. The main notebook is `module/notebooks/metabo_diet_harmonization.ipynb`.

Guide-to-notebook sequence

1. Read the matching lesson in this guide.
2. Open the notebook section with the same NB-L1 through NB-L5 key.
3. Run each code cell after a Run now heading, from top to bottom.
4. At a Learner edit heading, change only the named value or response, then run the cell.
5. Compare the output with the stated result and stop if a check fails.
6. Write the requested answer and use the Ready to move on? note before continuing.

Guide lesson	Notebook key	Run or complete
Lesson 1 - Why harmonization matters	NB-L1	NB-SETUP; locked configuration; scientific boundary
Lesson 2 - Comparing study design	NB-L2	Endpoint, sample-role, participant, and time audits
Lesson 3 - Harmonizing metadata/metabolites	NB-L3	NB-L3-CROSSWALK; labeled-standard and trace exercises
Lesson 4 - Guided analysis	NB-L4	NB-L4-CLASS; NB-L4-PCA-DIET; NB-L4-PCA-EXERCISE
Lesson 5 - Access transfer	NB-L5	Transfer-decision learner edit; NB-REPRO; posttest

Release control	Value
Diet case study	ST001521 - FARMM diet-anchored plasma study
Exercise case study	ST003348 - race-walking exercise-anchored serum study
Analysis boundary	Quantitative exploration stays within study; cross-study pooling is not authorized.
Offline path	Use the versioned cache and record the fallback in the source log.
Visual accessibility	Every proposed figure is paired with an equivalent text description in its lesson.

Module learning objectives

- LO1 - Explain how study design, specimen, time, and phenotype context shape interpretation.
- LO2 - Judge direct, partial, and non-comparability for a stated purpose.
- LO3 - Build a provenance-preserving RefMet crosswalk and document mapping uncertainty.
- LO4 - Retrieve, validate, harmonize, and explore public MW data reproducibly.
- LO5 - Verify access requirements and transfer the workflow safely to other resources.

Before you begin: first-time setup

The scientific material is written for learners who can read a data table and a scatterplot. You do not need prior experience with Metabolomics Workbench, RefMet, a terminal, Jupyter, or R. This is not an introductory Python course, and you will not be asked to write an analysis from a blank page.

The 153-minute course estimate begins after software setup. If Python, metabolomics, or principal component analysis (PCA) is new to you, allow an additional 30 to 60 minutes and keep the glossary open.

Choose your starting point

If you received the packaged course, keep these files together in one working folder:

1. [metabo_diet_learner_guide.pdf](#) - the lessons, pretest, posttest, and worked PCA figures.
2. [metabo_diet_analysis_bundle.zip](#) - the Python notebook, R companion, scripts, and public cached data.
3. [metabo_diet_templates.zip](#) - editable learner worksheets.

Extract both ZIP files before starting. If you are reading this page inside an extracted analysis bundle, that bundle is already extracted; extract only [module/support/metabo_diet_templates.zip](#).

If you cloned the repository, use its existing [module/](#) directory. Open the guide in [module/support/](#), use the learner worksheets in [module/templates/](#), and do not extract the analysis bundle on top of the clone.

Do not run the notebook from inside a ZIP. Keep the [module/](#) folder intact because the notebook uses its [scripts/](#) and [data/](#) subfolders.

Install Python and open Jupyter

A terminal is the text-based app used to enter the commands below. On macOS, open **Terminal**. On Windows, open **PowerShell**. Open the terminal in the folder that contains [module/](#), then type each command without a leading prompt symbol such as `$` or `>`.

Install [Python 3.12](#). Python 3.11 is also supported; substitute [python3.11](#) or [py -3.11](#) if that is the version you installed.

macOS or Linux

```
python3.12 --version
python3.12 -m venv .venv
./venv/bin/python -m pip install --upgrade pip
./venv/bin/python -m pip install -r module/notebooks/requirements-dev.txt
./venv/bin/python -m jupyter lab module/notebooks/metabo_diet_harmonization.ipynb
```

Windows PowerShell

```
py -3.12 --version
py -3.12 -m venv .venv
.\venv\Scripts\python.exe -m pip install --upgrade pip
.\venv\Scripts\python.exe -m pip install -r module\notebooks\requirements-dev.txt
.\venv\Scripts\python.exe -m jupyter lab module\notebooks\metabo_diet_harmonization.ipynb
```

The version check should report Python 3.11 or 3.12. The package installation can take several minutes.

Jupyter should then open in a browser. If it asks for a kernel, choose **Python 3 (ipykernel)** from the `.venv` environment. The first notebook diagnostic must report `Virtual environment: active`.

Jupyter in one minute

- A notebook contains text cells and code cells. Click a code cell and press **Shift+Enter** to run it.
- `[*]` beside a cell means it is still running. A number such as `[1]` means it finished.
- The file contains saved example outputs. Before beginning Lesson 1, choose **Kernel > Restart Kernel**, then run the required cells yourself from top to bottom.
- Run every code cell that follows a **Run now** heading.
- At a **Learner edit** heading, change only the named ALL_CAPS value or the clearly marked response text, then run that cell.
- A red traceback means the cell failed. Keep the message, stop, and use the setup table below or ask the instructor for help. Do not skip a failed check.

Setup help

What you see	What to do
<code>command not found</code> or “not recognized”	Confirm Python 3.12 is installed. Try the Python 3.11 command only if its version check reports 3.11.
<code>No module named ...</code>	Repeat the package-install command with the <code>.venv</code> Python path shown above, then restart the kernel.
Jupyter uses a different Python	Launch Jupyter with the direct <code>.venv</code> command above, choose Kernel > Change Kernel > Python 3 (ipykernel) , and confirm that the first diagnostic reports <code>Virtual environment: active</code> .
A notebook assertion or count check fails	Read the message and stop. Check the accession, cache, and earlier cells instead of deleting rows or changing the expected value.
A live request fails	Use the default validated cache. Live access is optional.

Optional technical check

The following command runs every code cell without pausing for the learner activities. It is useful for testing an installation, but it does not complete the course. It writes a new notebook and leaves the original file unchanged.

macOS or Linux:

```
./venv/bin/python module/scripts/execute_notebook.py --output metabo_diet_smoke_test.ipynb
```

Windows PowerShell:

```
.\venv\Scripts\python.exe module\scripts\execute_notebook.py --output metabo_diet_smoke_test.ipynb
```

R is optional

The Python notebook is the main activity. To read the R version without installing anything, open [module/notebooks/metabo_diet_R_appendix.html](#) in a browser.

To rerun the R companion, install R 4.3 or later plus RStudio or another Pandoc installation. First confirm Pandoc is available:

```
Rscript -e 'stopifnot(rmarkdown::pandoc_available()); rmarkdown::pandoc_version()'
```

Then run [Rscript module/notebooks/install_r_packages.R](#) and render the R Markdown file as described in [module/README.md](#).

Continue to the pretest

After Jupyter opens successfully, return to the learner guide and complete the pretest. For each lesson, read the guide first, run the matching notebook section, write the requested answer in the named worksheet or learner-edit cell, and check the “Ready to move on?” note.

Learner pretest

Answer without consulting the lessons. This baseline does not determine eligibility. The delivery platform should hide answers and rationales until submission.

Estimated time: 5 minutes

1. Which information is essential for interpreting a metabolite value labeled post?

- ☐ A. Only the metabolite name
- ☐ B. Only the participant's age
- ☐ C. The anchor event, elapsed time, specimen, physiological state, and assay context
- ☐ D. Only whether the value is above the median

Response: _____ Confidence (1-5): _____

2. A repository lists 120 samples from 30 people collected at four visits. Which statement is correct?

- ☐ A. The study contains 120 independent participants.
- ☐ B. The repeated samples are correlated and participant linkage must be preserved.
- ☐ C. Visit can be ignored if metabolite names match.
- ☐ D. PCA automatically accounts for the repeated design.

Response: _____ Confidence (1-5): _____

3. Two studies report the same standardized metabolite name. What does this establish?

- ☐ A. A common candidate naming key, subject to annotation and context review
- ☐ B. A common quantitative scale
- ☐ C. A common causal intervention effect
- ☐ D. Identical specimens and preprocessing

Response: _____ Confidence (1-5): _____

4. What is RefMet's primary role in this module?

- ☐ A. Convert every peak area to concentration
- ☐ B. Provide standardized analytical-chemistry nomenclature for metabolite structures and species
- ☐ C. Correct all batch effects
- ☐ D. Determine participant access eligibility

Response: _____ Confidence (1-5): _____

5. A name lookup returns two plausible metabolite candidates and no additional structural evidence. What is the best next step?

- ☐ A. Choose the first candidate.
- ☐ B. Choose the candidate that maximizes study overlap.
- ☐ C. Preserve both candidates, flag the row for review, and exclude it from exact overlap until resolved.
- ☐ D. Delete the source label.

Response: _____ Confidence (1-5): _____

6. Which Metabolomics Workbench REST study output is designed to return samples and experimental factors?

- ☐ A. summary
- ☐ B. factors
- ☐ C. classification
- ☐ D. molfile

Response: _____ Confidence (1-5): _____

7. What does separation in an exploratory PCA prove?

- ☐ A. The intervention caused every difference.
- ☐ B. The displayed samples differ along major variance directions under the selected preprocessing; causes require additional evidence.
- ☐ C. Batch effects have been removed.
- ☐ D. The data are normally distributed.

Response: _____ Confidence (1-5): _____

8. What is the best way to determine whether a dataset is open, agreement-gated, or controlled?

- ☐ A. Use the program's access status from memory.
- ☐ B. Check current first-party documentation for the exact dataset, release, and intended action and record the date.
- ☐ C. Assume that a public landing page permits download.
- ☐ D. Assume all NIH data are unrestricted.

Response: _____ Confidence (1-5): _____

9. For broad dataset discovery, plasma and serum are best classified as:

- ☐ **A.** Identical matrices
- ☐ **B.** Partially comparable blood-derived fluids whose exact matrices must remain visible
- ☐ **C.** Always not comparable for any purpose
- ☐ **D.** Directly comparable after z-scoring

Response: _____ Confidence (1-5): _____

10. What is the safest default architecture for participant-level data in a controlled Researcher Workbench?

- ☐ **A.** Download the participant table to a personal laptop for speed.
- ☐ **B.** Run approved code in the governed workspace and export only outputs permitted by current policy.
- ☐ **C.** Put the data in a public module cache.
- ☐ **D.** Share credentials with collaborators.

Response: _____ Confidence (1-5): _____

Lesson 1 - Why phenotype-to-metabolome harmonization matters

Estimated time: 20 minutes

Bloom levels: Remember, understand

Module objectives addressed: LO1, LO2, LO5

Learning objectives

By the end of this lesson, you will be able to:

1. Explain why metabolite measurements are not interpretable without study-design, specimen, timepoint, and phenotype context.
2. Distinguish structural harmonization from statistical pooling.
3. Recognize open, registration- or agreement-gated, and controlled-compute access patterns.
4. Locate the two configurable Metabolomics Workbench case studies used throughout the module.

Lesson map

Activity	Minutes
Pretest, additional assessment time	5
The phenotype-to-metabolome problem	5
The Metabo-Diet workflow and access patterns	5
Worked triage activity	7
Knowledge check and reflection	3
Instructional subtotal, excluding pretest	20
Learner time with pretest	25

In the notebook (NB-L1): Complete NB-SETUP, then run the Lesson 1 configuration cell. Confirm ST001521, ST003348, and the retrieval mode. In Section 1 of the cohort-comparison worksheet, record why the two quantitative matrices must stay separate.

Before you begin

Complete the pretest before opening the answer key. Its purpose is to establish a baseline, not to determine eligibility. Record your confidence separately from correctness; confidence and knowledge can change at different rates.

Three terms appear throughout the module. **Harmonization** aligns records while keeping important differences visible. **Pooling** places measurements from separate studies into one analysis. **Provenance** records where a value came from and how it changed. The glossary defines these and other unfamiliar terms.

This module keeps two configuration labels so a study pair can be replaced without rewriting the workflow. The release pair is now locked:

- `DIET_ACCESSION = ST001 521`: the public, human FARM diet-anchored plasma study.
- `EXERCISE_ACCESSION = ST003 348`: the public, human race-walking exercise-anchored serum study.

The variable names are not accession numbers and must never be sent to the REST API literally. The notebook configuration cell resolves them to the values above. The versioned source record is `module/data/provenance.json`. Before teaching, the instructor confirms that both studies remain public and that this record still matches the retrieved data.

1. A metabolite value is an observation in context

Imagine two rows labeled `glucose`. One value was measured in fasting plasma before a controlled diet; the other was measured in serum immediately after a maximal exercise bout. The analyte labels look compatible, but the observations are not interchangeable.

At least five layers shape what a metabolite value means:

1. **Biological question:** What exposure, intervention, or phenotype anchors the study?
2. **Sampling context:** Which specimen was collected, from whom, and under what conditions?
3. **Time meaning:** Does a label such as `post` mean after a meal, after weeks of feeding, immediately after exercise, or during recovery?
4. **Analytical process:** Which platform, method, batch structure, preprocessing, and units produced the value?
5. **Data representation:** How are samples, factors, metabolites, missing values, and identifiers encoded?

Harmonization makes these layers explicit. It does not erase them.

Keep this distinction: Structural harmonization can make records easier to compare. It does not, by itself, make participants, interventions, specimens, timepoints, platforms, units, or quantitative values exchangeable.

This distinction protects against a tempting but invalid workflow: matching metabolite names, stacking two intensity matrices, and interpreting the first principal component as a diet-versus-exercise biological effect. If study and phenotype are perfectly confounded, the observed separation may arise from

platform, sample handling, units, batch, population, or other design differences. No statistical technique can identify a unique biological cause from that contrast alone.

2. What harmonization can accomplish

The Metabo-Diet workflow supports several defensible tasks:

- Compare how studies represent intervention, specimen, timepoint, and phenotype variables.
- Standardize metabolite labels while preserving the submitted names and mapping evidence.
- Identify metabolites reported in both studies at a compatible annotation level.
- Describe where metadata concepts align, partially align, or do not align.
- Conduct exploratory analyses within each study using study-appropriate preprocessing.
- Compare patterns descriptively when the measurement and design context supports the comparison.
- Document unresolved mappings and incompatibilities for future reuse.

It does **not** automatically support:

- Direct comparison of raw or normalized intensities across platforms.
- Treating [post](#) timepoints as equivalent without physiological definitions.
- Treating plasma and serum as the same specimen.
- Treating a metabolite class label as a uniquely identified molecular structure.
- Inferring that an exposure caused a cross-study difference.
- Increasing effective sample size by concatenating unrelated cohorts.

Sometimes the correct decision is to keep a field or analyte separate. Record the reason so another analyst can review it.

3. The resources in this module

Metabolomics Workbench and NMDR

The Metabolomics Workbench hosts the National Metabolomics Data Repository (NMDR). Its REST service can return study summaries, factors, analyses, metabolite annotations, measurement tables, and mwTab records. The exercises use public Metabolomics Workbench studies. Because the accessions are stored as configuration values, they can be updated if a study changes or is replaced.

The mwTab representation organizes information into blocks. Depending on the record, these include study, subject, sample-factor, collection, sample-preparation, chromatography, mass-spectrometry or NMR, and metabolite-data content. The exact fields present vary across deposits. That variation is not an inconvenience to hide; it is evidence to inspect.

RefMet

RefMet provides an analytical-chemistry-centered reference nomenclature for metabolite structures and metabolite species. It is especially useful when deposited names differ in punctuation, synonym choice, or lipid notation. A RefMet match is evidence about nomenclature, not proof that two measurement features have identical analytical specificity. Lesson 3 develops this distinction.

MoTrPAC and Nutrition for Precision Health

The Molecular Transducers of Physical Activity Consortium (MoTrPAC) and Nutrition for Precision Health (NPH), powered by All of Us, appear as transfer cases rather than learner data dependencies. Their study designs show how exercise and diet phenotypes can be captured at consortium scale.

Access status is a property of a **specific dataset, release, and intended action**, not a permanent label attached to an entire program. Current MoTrPAC documentation states that public releases can be accessed without an account, while restricted or embargoed data can require authentication and a data-use agreement. NPH documentation states that data are stored and analyzed in the All of Us Researcher Workbench, which is available to approved researchers. Requirements can change, so Lesson 5 teaches verification from current first-party documentation.

4. Three access patterns

Use the following patterns as planning aids, not legal categories:

Pattern	Typical learner experience	Questions to verify
Open public retrieval	A public study can be viewed or retrieved without personal credentials.	Is this exact accession public? Is there an embargo? Are redistribution terms stated?
Account, agreement, or approval gated	Access requires identity, an account, acceptance of terms, a DUA, or review.	Who may apply? What uses and exports are allowed? How long might approval take?
Controlled compute	Participant-level data remain inside an approved environment; only permitted outputs leave it.	What tier is needed? What institutional and training requirements apply? What disclosure review or export rules govern outputs?

One resource may expose different datasets under different patterns. A public landing page does not prove that the desired data are public. Conversely, the existence of restricted data does not prove that all releases require registration.

Accessible description of the access-pattern figure

Picture three workspaces arranged from left to right. The first has an open door and a download arrow. The second has a sign-in badge and an agreement document. The third has data locked inside a cloud

workspace, with only a reviewed summary moving outward. A study can move between workspaces as its release status changes, so a calendar and version label appear below all three.

5. Worked example: a quick evidence check

This example introduces the checks for the release pair. Lesson 2 examines the dated study records in detail.

Claim to evaluate: "The diet and exercise datasets are both on Metabolomics Workbench, so their metabolite values can be pooled."

1. **Repository check:** Both are intended to be public MW studies. This supports a common retrieval mechanism, not common measurement properties.
2. **Identifier check:** Submitted names may be mapped to RefMet. This supports a documented analyte-name bridge, subject to annotation resolution.
3. **Specimen check:** Compare the sample source and collection protocol. Plasma versus serum, or fasting versus fed, can alter interpretation.
4. **Timepoint check:** Define each timepoint physiologically. A shared word such as [baseline](#) is insufficient.
5. **Platform and scale check:** Determine targeted versus untargeted analysis, chromatography, ion mode, quantification type, units, normalization, and batch structure.
6. **Design check:** Ask whether study and phenotype are separable. With one diet study and one exercise study, study membership is confounded with phenotype anchor.

Decision: Reject automatic quantitative pooling. Proceed with metadata comparison, identifier crosswalking, overlap description, and within-study exploratory analysis. Consider cross-study quantitative modeling only if a later, explicit compatibility assessment and design justify it.

6. Hands-on activity: open the source records

Open the learner cohort-comparison worksheet and complete the first four fields for both configured accessions.

1. Record [DIET_ACCESSION = ST001 521](#) and [EXERCISE_ACCESSION = ST003 348](#) from [module/data/provenance.json](#).
2. Record the diet [study page](#) and [REST summary](#), then the exercise [study page](#) and [REST summary](#).
3. Record the date and time you verified public availability.
4. Label the access evidence as [verified](#), [not verified](#), or [conflicting](#).
5. Write one sentence describing what public access does and does not imply about scientific comparability.

Stop condition: If either configured accession differs from the release manifest or its public release cannot be verified, do not substitute another study silently. Continue with the versioned cache if available, mark the source as cached, and report the discrepancy to the instructor.

7. Keep these distinctions

- Repository co-location supports common discovery and retrieval, not common measurement.
- A RefMet match supports a name-level bridge, not a common concentration or identification level.
- A broad specimen category must not erase plasma-versus-serum or other matrix differences.
- Physiological time must be defined before labels such as [baseline](#) or [post](#) are aligned.
- Public access does not establish scientific compatibility.
- Study-confounded separation must not be labeled a diet-versus-exercise effect.
- An unresolved or incompatible decision is a valid harmonization result.

8. Knowledge check

Answer before viewing the rationale in [module/assessments/knowledge_checks.json](#).

KC1-01. Two studies report a RefMet match for the same metabolite. Which conclusion is justified?

- A. Their quantitative values may be concatenated without further checks.
- B. Their submitted labels have a shared standardized-name candidate; analytical specificity and context still require review.
- C. The metabolite was confirmed by an authentic standard in both studies.
- D. Any between-study difference is biological.

KC1-02. Which evidence establishes the access pattern for a candidate dataset?

- A. The reputation of the research consortium.
- B. The access tier remembered from a prior release.
- C. Current first-party documentation and a dated check of the exact dataset and intended action.
- D. The fact that a publication cites the dataset.

KC1-03. When phenotype anchor and study are perfectly confounded, what is the safest interpretation of a PCA separation between studies?

- A. It proves diet and exercise have different metabolic effects.
- B. It proves the exercise cohort is more heterogeneous.
- C. It is exploratory separation that may reflect biological, technical, or design differences that cannot be uniquely attributed.
- D. It validates quantitative pooling.

9. Reflection

Complete this sentence in one or two lines:

A cross-study comparison would be useful for my work if it helped me _____, but it would become misleading if I _____.

Keep the response. You will revisit it after the analysis lesson.

Before you continue

Harmonization helps you compare records without erasing important differences. In Lesson 2, you will use that idea to compare the two study designs.

Primary sources and first-party documentation

1. Metabolomics Workbench. [REST Service, version 1.2](#). Updated July 22, 2025; accessed August 10, 2026.
2. Metabolomics Workbench. [mwTab file specification and repository tutorials](#). Accessed August 10, 2026.
3. Sud M, Fahy E, Cotter D, et al. [Metabolomics Workbench: an international repository for metabolomics data and metadata, metabolite standards, protocols, tutorials and training, and analysis tools](#). *Nucleic Acids Research*. 2016;44(D1):D463-D470.
4. Fahy E, Subramaniam S. [RefMet: a reference nomenclature for metabolomics](#). *Nature Methods*. 2020;17:1173-1174.
5. MoTrPAC. [Data access FAQ](#) and [Data Hub documentation](#). Accessed August 17, 2026.
6. NIH Common Fund. [Nutrition for Precision Health frequently asked questions](#). Accessed August 10, 2026.
7. Wilkinson MD, Dumontier M, Aalbersberg IJ, et al. [The FAIR Guiding Principles for scientific data management and stewardship](#). *Scientific Data*. 2016;3:160018.

Lesson 2 - Comparing study design and phenotype capture

Estimated time: 30 minutes

Bloom levels: Understand, analyze

Module objectives addressed: LO1, LO2

Learning objectives

By the end of this lesson, you will be able to:

1. Extract design, specimen, timepoint, phenotype, and provenance evidence from two Metabolomics Workbench records.
2. Classify fields as directly comparable, partially comparable, or not comparable for a stated purpose.
3. Explain how diet and exercise phenotypes acquire meaning from protocol details.
4. Use MoTrPAC and NPH as design references without treating their data as module dependencies.

Lesson map

Activity	Minutes
From labels to estimands	6
Five-layer study comparison	8
Consortium design references	4
Cohort-comparison worksheet	9
Knowledge check	3

In the notebook (NB-L2): Run the source, endpoint-size, biological-sample, and timepoint cells in order. Use the learner-edit cell to inspect one study, then enter the results in Sections 3, 5, 6, and 9 of the cohort-comparison worksheet.

1. Begin with the question, not the column name

A variable is comparable only relative to a purpose. Age in years may be sufficiently comparable for a descriptive cohort table but insufficient if one study reports exact age and another reports broad, top-coded categories. A [post](#) sample may be suitable for identifying that both studies collected after an intervention but unsuitable for modeling a common acute-response time.

Before aligning columns, write the target question as an **estimand sketch**:

- **Population:** Who is represented?

- **Exposure or intervention:** What contrast is of interest?
- **Outcome:** Which metabolite measure or derived quantity is being interpreted?
- **Time:** At what biological interval is the outcome defined?
- **Summary:** Is the target a within-person change, a group mean, a rank, an overlap count, or something else?

For this module, the safe default target is descriptive:

Describe which metabolite identities and broad chemical classes are represented in both studies, while preserving differences in population, specimen, timepoint, platform, and measurement scale.

That target supports crosswalking and overlap summaries. It does not require the two cohorts to estimate the same intervention effect.

2. The five-layer comparison

Use the cohort-comparison worksheet to collect evidence at five layers. For every entry, capture the source location and avoid filling a blank by inference.

Layer A - Study design and population

Record:

- Observational versus interventional design.
- Parallel, crossover, repeated-measures, or other structure.
- Recruitment setting and inclusion or exclusion criteria.
- Unit of assignment and unit of analysis.
- Number of participants and samples, keeping those counts distinct.
- Randomization, masking, washout, run-in, or familiarization when applicable.
- Attrition and missing visits if reported.

The number of samples is not the number of independent participants. Repeated samples from one participant are correlated. If participant identifiers are absent or cannot be linked reliably, within-person analyses may be impossible even when multiple timepoints exist.

Layer B - Phenotype anchor

For a diet-anchored study, inspect:

- Food or dietary pattern definition.
- Controlled feeding versus advice or free-living exposure.
- Energy balance, meal timing, compliance measures, and washout.
- Fasting state and any test-meal challenge.

- Dietary assessment method and its reference period.

For an exercise-anchored study, inspect:

- Modality, intensity, duration, dose, and supervision.
- Acute bout versus chronic training.
- Fitness or performance measures.
- Pre-exercise standardization, recent activity restrictions, and feeding state.
- Recovery intervals and workload scaling.

The words **diet** and **exercise** are topic labels, not exposure definitions. The worksheet should contain protocol-level details sufficient for another analyst to understand the contrast.

Layer C - Specimen and pre-analytical context

Record:

- Biospecimen, matrix, and anatomical source.
- Collection tube or anticoagulant when relevant.
- Fasting or fed status.
- Collection clock time if available.
- Processing delay, centrifugation, aliquoting, and storage temperature.
- Freeze-thaw information if available.

Plasma and serum are both blood-derived, but they are not identical matrices. Treat them as partially comparable at most, and only for a purpose that tolerates the distinction. A field left unreported is **not reported**, not **same**.

Layer D - Time semantics

Translate each deposited time label into a physiological definition. Use a canonical structure such as:

anchor | offset | duration/unit | state | visit/order

Examples:

- **exercise_bout_end | +0 | min | acute_response | visit_2**
- **exercise_bout_end | +180 | min | recovery | visit_2**
- **diet_period_start | +14 | day | end_of_feeding_period | period_1**
- **test_meal_start | -10 | min | fasting_baseline | visit_1**

Do not translate **post** without evidence. When the necessary reference point or offset is absent, flag the timepoint as unresolved.

Layer E - Assay and metadata provenance

Record:

- Analysis type, platform, chromatography, ion mode, and targeted or untargeted status.
- Quantification or abundance scale and units.
- Normalization, transformation, blank correction, batch correction, and missing-value handling when documented.
- Analysis identifier as well as study accession.
- mwTab block or REST response that supports each field.
- Retrieval date, endpoint, and record version or checksum when available.

A study may contain multiple analyses. Never assume a study-level description applies identically to every analysis table.

3. Comparability is a reasoned decision

Use these operational categories:

Directly comparable

The two fields represent the same construct at sufficient granularity for the stated use, with compatible coding or a transparent lossless conversion.

Example: Both studies report participant age in years at enrollment, and the target is a descriptive median and range. Verify population definitions and disclosure rules before reporting.

Partially comparable

The fields share a broader construct but differ in granularity, timing, protocol, matrix, or measurement. They may support stratified description or a coarser derived category, but the difference must remain visible.

Example: One study records exact fasting duration; another records only [fasted](#). Both can support a high-level fasting-state flag, but not a dose-response model by fasting hours.

Not comparable

The fields represent different constructs, lack essential provenance, or would require an unsupported assumption.

Example: [post_diet](#) after a 14-day feeding period and [post_exercise](#) immediately after an acute bout cannot be treated as one common post-intervention timepoint.

Not yet assessable

Use this fourth workflow status when evidence is missing or the accession is unresolved. It prevents uncertainty from being silently converted into incompatibility or compatibility. Before analysis, resolve it or explicitly exclude it.

4. Worked example with the locked configuration

The following decisions are grounded in the locked Metabolomics Workbench records and remain subject to the dated verification recorded in your worksheet.

Domain	DIET_ACCESSION = ST001521	EXERCISE_ACCESSION = ST003348	Decision for this module
Species	Human in summary and subject metadata	Human in summary and subject metadata	Direct at a broad species level
Matrix	Blood plasma	Collection metadata specifies blood serum, although factors say blood	Partial for discovery; keep plasma and serum distinct
Time	Baseline and Days 5, 9, 12, and 15 across feeding plus antibiotic/PEG procedures	REST, immediate post-exercise, 3-hour recovery, and 22-hour recovery	Not a shared time axis
Phenotype	Three diet groups in a longitudinal FARM protocol	Acute endurance race-walking and recovery	Not one common intervention variable
Metabolite name	Submitted name and RefMet output	Submitted name and RefMet output	Exact-name comparison possible after Lesson 3 review
Intensity	Four LC-MS analyses, unitless peak areas	Two LC-MS analyses, peak area	No cross-study quantitative pooling

It is fine to mark a row [not yet assessable](#) when the source does not provide enough information.

5. Consortium-scale design references

MoTrPAC

MoTrPAC's published human protocol describes a multicenter effort designed to characterize molecular responses to acute exercise and exercise training. It includes structured exercise phenotyping and collection of blood, skeletal muscle, and adipose tissue in defined protocol contexts. Use the protocol as an example of how modality, dose, tissue, and time are specified. Do not copy its timepoint labels onto [EXERCISE_ACCESSION](#); retrieve the actual MW study protocol and factors.

Nutrition for Precision Health

NIH describes NPH as a modular study nested in All of Us. Public program documentation distinguishes a larger baseline/test-meal module from free-living and domiciled controlled-feeding modules. Use it as

an example of how dietary exposure, test meals, and multimodal measures can be organized. Do not infer that [DIET_ACCESSION](#) follows the NPH design.

These examples teach **design transfer**: recognize a structure, then verify whether it exists in the target data. They do not supply missing metadata for the MW case studies.

Accessible description of the comparison diagram

The diagram has two vertical lanes. The diet lane moves from diet assignment to adherence and feeding state, then to specimen collection. The exercise lane moves from exercise modality and dose to exertion and recovery, then to specimen collection. Both lanes feed into a shared assay box, but each arrow carries its own time, specimen, and protocol tags. A dashed line between the lanes is labeled [compare metadata](#); there is no arrow labeled [pool values](#).

6. Hands-on activity: complete the cohort-comparison worksheet

Use [module/templates/cohort_comparison_worksheet_learner.md](#).

For the timed course, complete these eight rows in the comparability matrix: species, specific matrix, timepoint, participant count, repeated-measures structure, phenotype anchor, units/scale, and platform/mode. Use the larger worksheet as a project extension after the lesson. A first-time learner is not expected to fill every blank table cell in 30 minutes.

Step 1 - Resolve and document

For [DIET_ACCESSION = ST001 521](#) and [EXERCISE_ACCESSION = ST003 348](#), record:

- Resolved study accession and all analysis identifiers used.
- Landing page, REST endpoints, retrieval timestamp, and public-release evidence.
- Publication or protocol link if the record provides one.

If live retrieval fails, use only the versioned cached record supplied with the module, retain the configuration key in every source field, and mark the retrieval source as cached.

Step 2 - Extract independently

Complete each study column without looking at the other study. This reduces the temptation to translate a field into whatever appears in the comparison study.

For every cell, use one of:

- A value plus source location.
- [NR](#) for not reported after a documented search.
- [NA](#) for not applicable.
- [UNRESOLVED](#) when the accession, analysis selection, or meaning is pending.

Step 3 - State the purpose

Choose one comparison purpose:

1. Metadata inventory.
2. Metabolite-name overlap.
3. Class-level presence or coverage.
4. Within-study change followed by qualitative pattern comparison.

Do not select pooled quantitative inference for this exercise.

Step 4 - Classify and justify

For at least eight rows, mark directly comparable, partially comparable, not comparable, or not yet assessable. Write a one-sentence evidence-based justification. A justification should name the relevant similarity or difference rather than merely repeat the category.

Step 5 - Write an exclusion statement

Complete:

We will not combine _____ across the two studies because _____. We can still compare _____ by _____.

Completion criteria

Your worksheet is complete when:

- Every substantive entry has a source or an explicit missingness code.
- Participant and sample counts are not conflated.
- Timepoints have physiological anchors or are flagged.
- Assay identifiers and units are recorded.
- At least one field is intentionally not harmonized.

7. Comparison rules

- A shared label is a hypothesis of equivalence, not proof.
- Missing metadata does not mean the protocols were identical.
- Recoding to a broader category trades detail for compatibility; document the loss.
- A crossover period, repeated sample, or technical replicate is not an independent participant.
- Do not use consortium documentation to fill gaps in a different MW study.
- Do not select only the metadata that support a desired integration.
- Record the analysis identifier because one accession may contain multiple assays.

8. Knowledge check

KC2-01. One study records fasting duration in hours and another records only yes/no fasting status. For a binary fasted-versus-not-fasted description, how should the fields be classified?

- A. Directly comparable with no documentation.
- B. Partially comparable after a documented derivation and loss of detail.
- C. Not comparable for any purpose.
- D. Directly comparable for modeling fasting duration.

KC2-02. What is the best evidence that two rows at three timepoints come from the same participant?

- A. Similar metabolite profiles.
- B. Adjacent sample identifiers.
- C. A documented subject-sample relationship in the deposited metadata.
- D. The word [longitudinal](#) in the publication title.

KC2-03. Why is [post](#) insufficient as a harmonized timepoint?

- A. Post-intervention samples are never useful.
- B. It lacks an intervention anchor, offset, and physiological state.
- C. It cannot appear in mwTab.
- D. It always means immediate recovery.

Before you start the crosswalk

Decide what you want to compare first. Then extract each record on its own terms and explain why each field is or is not suitable for that purpose. Lesson 3 applies those decisions to the metabolite crosswalk.

Primary sources and first-party documentation

1. Metabolomics Workbench. [mwTab file specification, version 1.7, and tutorials](#). Accessed August 10, 2026.
2. Metabolomics Workbench. [REST study-context endpoints](#). Accessed August 10, 2026.
3. MoTrPAC Study Group, Jakicic JM, Kohrt WM, et al. [Molecular Transducers of Physical Activity Consortium \(MoTrPAC\): human studies design and protocol](#). *Journal of Applied Physiology*. 2024;137(3):473-493.
4. MoTrPAC. [Adult Study Protocol, version 3.0](#). August 22, 2023.
5. NIH Common Fund. [Nutrition for Precision Health](#) and [program FAQ](#). Accessed August 10, 2026.
6. All of Us Research Program. [Researcher Workbench](#). Accessed August 10, 2026.

Lesson 3 - Harmonizing metabolomics and metadata

Estimated time: 35 minutes

Bloom level: Apply

Module objectives addressed: LO2, LO3

Learning objectives

By the end of this lesson, you will be able to:

1. Build a provenance-preserving crosswalk from submitted metabolite labels to RefMet candidates.
2. Distinguish a naming match from evidence of chemical identification and analytical equivalence.
3. Align specimen, timepoint, variable, and unit metadata without concealing information loss.
4. Flag ambiguous or unresolved mappings instead of forcing a one-to-one result.
5. Evaluate whether a crosswalk is reproducible enough for downstream use.

The locked configuration is `DIET_ACCESSION = ST001521` and `EXERCISE_ACCESSION = ST003348`. Continue using the variable names in the crosswalk so the source accession remains explicit and a fallback pair can be substituted transparently.

Lesson map

Activity	Minutes
Identifier problem and RefMet	8
mwTab and provenance	5
Specimen, time, variables, and units	7
Crosswalk activity	12
Knowledge check	3

In the notebook (NB-L3): Run NB-L3-CROSSWALK and inspect the labeled-standard table. Confirm the 153 raw and 145 conservative counts. Then trace one retained RefMet name and record what the match does and does not show in Tab A of the crosswalk.

1. Identifier systems answer different questions

Metabolomics records can contain common names, vendor labels, abbreviations, database identifiers, formulas, masses, or structural keys. These are not interchangeable fields.

Field	What it can help represent	Important limitation
Submitted name	What the depositor reported	Synonyms, punctuation, adducts, and uncertain annotation can vary
RefMet name	Standardized analytical-chemistry nomenclature for a structure or metabolite species	A name match does not prove the underlying feature identification or equal structural resolution
InChIKey	Hash derived from a chemical structure representation	Layers encode different degrees of specificity; salts, protonation, tautomer, and stereochemical representation can differ
PubChem CID	Record identifier in PubChem	Different records may reflect different forms or levels of structural detail
HMDB ID	Record identifier in the Human Metabolome Database	Coverage and record relationships differ from other resources
KEGG compound ID	Identifier in KEGG's compound representation	Useful for pathway linking but not a universal metabolite identity key
Formula or exact mass	Composition or mass evidence	Isomers can share formula or mass; mass alone is not an identification

RefMet provides the common naming field in this crosswalk. It covers discrete structures and analytically reported species, including lipid sum compositions. Keep the submitted labels and other identifiers beside it; a standardized name should not replace the original evidence.

2. Annotation resolution must be preserved

Suppose one study reports [PC 34:1](#) and another reports a phosphatidylcholine with two specified acyl chains. Even if a database relationship connects them, the annotations have different structural resolution. Collapsing both to the more specific structure would manufacture information. Collapsing to a broader class may be defensible for a class-level presence summary, but that is a derived analysis decision that must be recorded.

For each candidate match, record an **annotation-resolution status** such as:

- [discrete_structure](#)
- [stereochemistry_unspecified](#)
- [lipid_molecular_species](#)
- [lipid_sum_composition](#)
- [compound_class_only](#)
- [unknown](#)

Also record the identification evidence reported by the study when available. The Metabolomics Standards Initiative distinguishes levels of identification confidence; a standardized label does not upgrade that confidence.

Mapping limit: Normalize a label only to the level supported by the deposited analytical evidence. Never infer a specific isomer, stereoisomer, lipid chain arrangement, or identification level from a fuzzy name match.

3. A reproducible mapping workflow

Use the following sequence for every distinct submitted analyte label.

Step 1 - Preserve source identity

Record:

- Study accession: `DIET_ACCESSION` or `EXERCISE_ACCESSION`, resolved in the release manifest.
- Analysis identifier.
- Original analyte label exactly as deposited.
- Original metabolite or feature identifier.
- Original database identifiers and units.
- Source endpoint, retrieval timestamp, and source-table location.

Never overwrite the original label.

Step 2 - Apply transparent normalization for lookup

Create a separate lookup string. Permissible preprocessing may include trimming whitespace or standardizing obvious delimiter characters. Record each transformation. Do not remove stereochemical markers, lipid chain details, charge, adduct, isotope label, or derivatization information merely to obtain a match.

Step 3 - Query RefMet

The documented REST pattern for matching a name is:

`https://www.metabolomicsworkbench.org/rest/refmet/match/{URL_ENCODED_NAME}/name/`

The exact response must be captured by the module notebook. A manual web conversion is acceptable for a small exercise, but record the query date and string.

Step 4 - Validate the candidate

Compare all available evidence:

- Submitted and standardized names.
- Formula and mass.

- InChIKey or external IDs.
- Annotation resolution.
- Analytical method and identification evidence.
- Whether the candidate represents an adduct, derivative, isotope-labeled standard, or unresolved mixture.

The absence of a contradiction is not proof of identity. Use the confidence and decision fields.

Step 5 - Assign mapping status

Recommended statuses are:

- **accepted_exact**: evidence supports the same standardized name and resolution.
- **accepted_broader**: a broader representation is intentionally used for a stated analysis.
- **review_required**: more than one candidate or conflicting evidence remains.
- **unmapped**: no supported RefMet candidate was found.
- **excluded_nonbiological**: internal standard, contaminant, adduct-only label, or other intentionally excluded feature, with a reason.
- **not_evaluated**: mapping has not been reviewed.

Automated output should begin as **not_evaluated** or **review_required**, not **accepted_exact**, unless the validation rule and evidence are explicit.

Step 6 - Preserve the decision

Record reviewer, date, rule or evidence, decision reason, and downstream eligibility. If a decision changes, append a new record or version the table; do not erase the earlier rationale.

Accessible description of the mapping workflow

Imagine a six-column ledger read from left to right. The deposited feature and exact label stay fixed in the first column. A separate lookup string and RefMet response occupy the next two columns. Evidence and annotation resolution appear in the fourth column. A human or versioned rule records an accepted, broader, review-required, unmapped, or excluded decision in the fifth. The final column contains separate yes/no/review flags for exact-name overlap, class summary, within-study analysis, and cross-study analysis. Arrows never replace the first column; they add traceable decisions beside it.

4. Worked mapping example

These source and RefMet strings appear in the current public metabolite responses. The review decisions are instructional and must be regenerated if the repository record changes.

Source/analysis	Deposited label	Source RefMet string	Evidence question	Safe decision
ST001 521, AN002533	2-Aminooctanoic acid	2-Aminocaprylic acid	Do synonym and identifier evidence support the same structure?	<code>review_required</code> , then potentially <code>accepted_exact</code>
ST001 521, AN002533	LPE(18:0)_A	LPE 18:0	Does _A distinguish a separate source feature, and is sum composition the supported resolution?	Preserve feature; accepted set key only after review
ST001 521, AN002533	LPC(P-18:0)/ LPC(O-18:1)_A	LPC P-18:0 or LPC O-18:1	Is the unresolved ether-lipid alternative preserved?	<code>accepted_broader</code> for class/species purposes, not a unique structure
ST003 348, AN005484	Mannitol+Sorbitol	Mannitol	Has a composite/isomeric label been narrowed without support?	<code>review_required</code>
ST003 348, AN005484	NegX-RT295MZ165	blank	Is this an unidentified feature?	<code>unmapped</code> ; do not name from mass alone
ST003 348, AN005484	Palmitic acid-[13C]16	Palmitic acid	Is the source an isotope-labeled analytical standard rather than endogenous palmitic acid?	<code>excluded_nonbiological</code> for biological overlap

The last row shows why a nonblank standardized field is not automatically eligible. In the audited build, eight ST003 348 isotope/internal-standard rows map to ordinary RefMet labels; excluding them reduces the raw exact-name overlap from 153 to a conservative biological overlap of 145.

5. mwTab as an evidence map

An mwTab record is sectioned rather than a single flat data frame. Typical content includes project and study descriptions; subject and sample factors; collection and sample-preparation protocols; chromatography and instrument metadata; and a metabolite-data block. Presence and completeness vary by record and analysis.

For the configured studies, retrieve at least the following after substituting each variable's locked value:

```
/rest/study/study_id/{DIET_ACCESSION}/summary
/rest/study/study_id/{DIET_ACCESSION}/factors
/rest/study/study_id/{DIET_ACCESSION}/analysis
/rest/study/study_id/{DIET_ACCESSION}/metabolites
/rest/study/study_id/{DIET_ACCESSION}/data
```

Repeat with {EXERCISE_ACCESSION}. Retrieve each selected analysis as mwTab when analysis-specific evidence is needed:

```
/rest/study/analysis_id/{ANALYSIS_ID}/mwtab
```

Do not assume the shape of a response from the endpoint name. Inspect the returned keys, status code, and content type. A successful HTTP response can still contain an unexpected schema or an error message encoded as text.

Provenance minimum

Every crosswalk row must be traceable to:

- Repository and base URL.
- Study accession and analysis identifier.
- Exact endpoint or source file.
- Retrieval timestamp in UTC.
- Original label and source identifier.
- Mapping service and query string.
- Mapping result and review decision.
- Reviewer or automated rule version.

These fields make the crosswalk reusable and support FAIR principles of findability, interoperability, and reuse.

6. Harmonizing specimen and time metadata

Metabolite-name mapping is only one part of the crosswalk.

Specimen crosswalk

Keep at least three fields:

1. `source_specimen_label`: exact deposited value.
2. `normalized_specimen_category`: a controlled broader category used for a specific purpose.
3. `specimen_compatibility_note`: pre-analytical differences and any information loss.

For example, plasma and serum could both receive a broad `blood_derived_fluid` category for discovery, while remaining distinct in the exact matrix field. That category does not authorize quantitative pooling.

Timepoint crosswalk

Keep the source label and derive structured fields only when evidence supports them:

- `anchor_event`
- `offset_value`
- `offset_unit`
- `physiological_state`
- `visit_or_period`
- `within_subject_order`
- `derivation_source`

If the label is [T2](#) and the protocol needed to decode it is missing, use [UNRESOLVED](#); do not guess.

Variable crosswalk

Separate:

- Source field and coding.
- Construct definition.
- Harmonized field and coding.
- Transformation rule.
- Information lost.
- Eligibility by analysis purpose.

For categorical variables, document reference levels and distinguish missing, unknown, not applicable, and structurally absent.

7. Unit reconciliation

Convert units only when all of the following are true:

1. The values measure the same quantity.
2. Units are known and convertible.
3. Specimen basis and denominator match or can be validly transformed.
4. Calibration and quantification type support the intended comparison.
5. The conversion formula is recorded and tested.

Examples of different situations:

- [umol/L](#) and [mmol/L](#) are dimensionally convertible for the same quantified analyte and compatible matrix, subject to the remaining checks.
- Peak area and [umol/L](#) are not made comparable by rescaling.
- Relative abundance normalized within one study and raw area in another are not a common measurement scale.
- Log-transformed and untransformed measurements can be reconciled only if the transform base, offset, and original scale are known and the comparison is otherwise valid.

Keep [source_value](#) and [source_unit](#) even after a justified conversion. Store the converted field separately.

8. Hands-on activity: build and audit the crosswalk

Use [module/templates/metabolite_metadata_crosswalk_learner.md](#).

For the timed course, complete three worked rows: one retained shared RefMet name, one isotope-labeled or internal-standard exclusion, and one ambiguous or unmapped example. The 12-row metabolite table and full metadata audit are a project extension. This smaller path lets a first-time learner practice each decision type before scaling up.

Part A - Metabolites

1. For the course path, select one retained shared name, one excluded standard, and one ambiguous or unmapped label. For the project extension, select at least six labels from each study.
2. Include at least one apparent overlap, one lipid label if present, one unmapped or unidentified feature if present, and one label requiring manual review. If a requested case does not occur, document that rather than fabricating it.
3. Preserve original labels and identifiers.
4. Run or inspect the RefMet query for each distinct label.
5. Assign mapping and annotation-resolution statuses.
6. Mark eligibility separately for exact-name overlap, broad-class summary, and quantitative comparison.

Part B - Metadata

Add at least:

- Two specimen fields.
- Three timepoint values.
- One participant or sample identifier relationship.
- One phenotype field.
- One unit or scale field.

Part C - Reproducibility audit

Exchange the crosswalk with a partner or use the self-audit prompts:

- Can another analyst find the exact source row?
- Can they reproduce the lookup string?
- Can they tell which output was automated and which was reviewed?
- Is information loss stated?
- Are unresolved cases preserved?
- Is downstream eligibility purpose-specific?
- Could the table be regenerated after a study update?

Flag any row that fails one of these checks. The goal is an auditable table, not a maximum mapping rate.

9. Mapping rules

- RefMet standardizes nomenclature; it does not retroactively validate compound identification.
- A one-to-many or many-to-one mapping is not automatically an error; it may reflect different resolution.
- Do not choose a candidate solely because it creates more overlap.
- Never convert unidentified features into named metabolites using mass alone.
- Preserve exact specimen and time labels alongside broader categories.
- Do not treat unit conversion as batch correction.
- Exclude internal standards and nonbiological features by explicit rule, not by silently deleting inconvenient rows.
- Version mapping decisions because repositories and reference databases evolve.

10. Knowledge check

KC3-01. A fuzzy RefMet lookup returns two plausible candidates and the deposited record has no formula or external identifier. What should you do?

- A. Select the first candidate.
- B. Select whichever candidate appears in the other study.
- C. Mark the row `review_required` or `unmapped`, preserve both candidates, and exclude it from exact overlap until resolved.
- D. Replace it with a chemical class label and call it exact.

KC3-02. When can `umol/L` be converted to `mmol/L` for cross-study comparison?

- A. Whenever both unit strings are present.
- B. Only after confirming the same quantity, compatible matrix and denominator, quantification basis, and a recorded conversion.
- C. Only for lipids.
- D. Never.

KC3-03. Which crosswalk design is most reproducible?

- A. One column containing the final clean name.
- B. Original and standardized names, source and query provenance, mapping status, evidence, reviewer, decision reason, and purpose-specific eligibility.
- C. A list containing only mapped analytes.
- D. A table in which all ambiguities are manually forced to one value.

What the crosswalk should preserve

A good crosswalk preserves the source label, the proposed standardized name, the supporting record, and the decision. Keep unresolved rows marked as unresolved so later analyses do not treat them as confirmed matches.

Primary sources and first-party documentation

1. Metabolomics Workbench. [RefMet database and name-conversion resources](#). Accessed August 10, 2026.
2. Metabolomics Workbench. [RefMet naming conventions](#). Accessed August 10, 2026.
3. Fahy E, Subramaniam S. [RefMet: a reference nomenclature for metabolomics](#). *Nature Methods*. 2020;17:1173-1174.
4. Metabolomics Workbench. [REST Service, version 1.2](#). Updated July 22, 2025; accessed August 10, 2026.
5. Metabolomics Workbench. [mwTab file specification and tutorials](#). Accessed August 10, 2026.
6. Powell CD, Moseley HNB. [The mwtab Python Library for RESTful access and enhanced quality control, deposition, and curation of the Metabolomics Workbench Data Repository](#). *Metabolites*. 2021;11(3):163.
7. Sumner LW, Amberg A, Barrett D, et al. [Proposed minimum reporting standards for chemical analysis](#). *Metabolomics*. 2007;3:211-221.

Lesson 4 - Guided analysis and biological interpretation

Estimated time: 40 minutes

Bloom levels: Apply, analyze, evaluate

Module objectives addressed: LO2, LO4

Learning objectives

By the end of this lesson, you will be able to:

1. Retrieve and validate study summaries, factors, analyses, metabolites, and data through the Metabolomics Workbench REST API.
2. Apply the reviewed crosswalk without losing study, analysis, sample, specimen, or timepoint provenance.
3. Quantify name-level overlap and summarize chemical classes with explicit eligibility rules.
4. Run and interpret principal component analysis within each study as an exploratory diagnostic.
5. Separate observations supported by the output from explanations that remain biologically or technically ambiguous.

Lesson map

Activity	Minutes
Configure, retrieve, and validate	8
Join and audit the crosswalk	6
Overlap and class summaries	7
Within-study PCA	9
Interpretation challenge	7
Knowledge check	3

In the notebook (NB-L4): Run NB-L4-CLASS, NB-L4-PCA-DIET, and NB-L4-PCA-EXERCISE in order. Complete both learner-edit cells and the four-sentence note before reading the sample answer. The same PCA figures appear in the appendix.

1. Locked case-study configuration

The release configuration is:

```
DIET_ACCESSION = ST001 521
EXERCISE_ACCESSION = ST003 348
```

Keep the variable names in code and prose even though the accessions are now locked. This preserves the proposal's one-cell substitution design and makes a fallback pair possible.

Diet case: [ST001521](#)

Metabolomics Workbench describes this FARM study as a longitudinal human feeding study with three diet groups and plasma metabolomics. The study record reports 30 volunteers and five biological collection times. The factor response contains 160 rows: 150 participant-timepoint rows plus 10 pooled quality-control rows labeled [QPP...](#) with diet, sex, and time recorded as [NA](#). Four analysis records cover different chromatography and ion-mode combinations and report unitless peak areas.

[ST001521](#) is not a simple three-arm diet trial. Participants also underwent antibiotic and polyethylene glycol procedures during the time course; the vegan group continued its usual diet as outpatients, while 20 omnivores were randomized to an omnivore diet or exclusive enteral nutrition in an inpatient unit. Changes across days therefore cannot be attributed to diet alone without a design-appropriate model and assumptions.

Exercise case: [ST003348](#)

Metabolomics Workbench describes an endurance exercise study of race walkers with 76 samples: 19 athletes at four timepoints. The project record defines [REST](#) as pre-exercise, [STAT](#) as immediately post-exercise, [REC3](#) as 3 hours into recovery, and [REC22](#) as 22 hours post-exercise. Two analysis records cover reversed-phase positive- and negative-ion modes and report peak area.

The factors endpoint uses the broad sample-source label [blood](#), while the collection record specifies [Blood \(serum\)](#) and describes clotting and serum preparation. Keep [blood](#) as the submitted factor value, use serum as the more specific matrix supported by the collection record, and cite both sources.

Cross-study compatibility decision

The diet study uses plasma; the exercise study uses serum. Their time axes also represent different processes: study days spanning feeding and microbiota-depletion procedures versus minutes to hours after an endurance exercise bout. Treat both domains as partially comparable for broad discovery and not directly comparable for a pooled quantitative effect.

2. Open the notebook safely

Open [module/notebooks/metabo_diet_harmonization.ipynb](#) in the **Python 3 (ipykernel)** kernel launched from [.venv](#). Complete [NB-SETUP](#) and run [NB-L1](#) through [NB-L3](#) before starting this lesson; the Lesson 4 cells depend on those validated objects. Then follow this order:

1. Run [NB-L4-CLASS](#) and confirm that the class counts sum to 145.
2. Run [NB-L4-PCA-DIET](#) and record its sample/feature counts and explained variance.

3. Run [NB-L4-PCA-EXERCISE](#) and record the same fields without comparing coordinate values across models.
4. Complete the shared-class and sensitivity learner-edit cells.
5. Write the bounded interpretation before revealing the sample answer.

Confirm the configuration cell shows the accessions above. The data provenance manifest at [module/data/provenance.json](#) should also provide:

- Selected analysis IDs or the rule used to include multiple analyses.
- Retrieval date and API base URL.
- Cached-file paths and checksums.
- Dependency versions.
- Crosswalk version.

Do not continue if the displayed accession differs from [module/data/provenance.json](#).

Live retrieval and cached fallback

For each accession, the notebook requests:

- [summary](#)
- [factors](#)
- [analysis](#)
- [metabolites](#)
- [data](#)

The release notebook defaults to the validated cache so every learner receives a deterministic run. Set [METABO_DIET_LIVE=1](#) to attempt live retrieval first. The live path falls back endpoint-by-endpoint only after recording:

- The failed endpoint and status.
- The cached retrieval timestamp.
- A checksum match.
- A visible source entry of [validated cache](#) in the input source log.

Cached data keep the lesson runnable; they do not establish current public availability. Record the outage and re-verify access later.

3. Validate before analyzing

An HTTP [200](#) is not sufficient validation. The notebook's audit table should answer:

1. Did every response parse in the expected format?
2. Does each record contain the requested study or analysis identifier?
3. Are sample identifiers unique at the expected level?
4. How many biological, pooled QC, blank, and unknown rows are present?
5. Do factor counts agree with the documented repeated-measures structure?
6. Are all matrix columns represented in the factor table?
7. Are metabolite labels unique after applying the chosen feature key?
8. Are units and analytical modes explicit?

Expected structural checks for this release

These are validation targets, not biological results:

Check	ST001521	ST003348
Factor rows	160	76
Biological participant-timepoint rows expected	150	76
Pooled QPP... rows expected	10	0 identified from the factor table
Biological participants described	30	19
Biological timepoints per participant	5	4
Analysis records	4	2
Matrix	Plasma	Serum, supported by collection metadata

If the live response differs, stop and inspect. A repository revision may be legitimate, but silently forcing current data to match old expectations defeats provenance tracking.

4. Separate analytical roles before filtering

Create a [sample_role](#) field with evidence. At minimum:

- [biological](#)
- [pooled_qc](#)
- [blank](#)
- [technical_standard](#)
- [unknown](#)

For [ST001521](#), rows with local identifiers beginning [QPP](#) and missing diet, sex, and time are pooled QC candidates supported by the record. The notebook should keep them for assay diagnostics but exclude

them from biological PCA and participant counts. Never discard QC rows before confirming their role; their clustering and drift can reveal analytical problems.

For [ST003 348](#), identifiers encode an apparent participant and time suffix, but the notebook must derive repeated-measures links using a documented rule and validate the expected 19 by 4 structure. A string pattern is a parsing hypothesis until confirmed against study metadata.

5. Join the reviewed crosswalk

Join within study and analysis first, using the most specific stable keys available. Avoid joining solely on submitted metabolite name because:

- The same name may occur in multiple analytical modes.
- Different features can map to the same broad RefMet name.
- Duplicate labels can create a many-to-many join and inflate rows.

After each join, report:

- Rows before and after.
- Matched, unmatched, and multiply matched source features.
- Duplicate join keys.
- Counts by mapping status and annotation resolution.
- Exclusions and reasons.

The learner should be able to trace any harmonized value back to accession, analysis, sample, and original feature.

Stop condition: If a join unexpectedly increases the number of measurement rows, diagnose the key cardinality before continuing. Do not remove duplicates merely to restore the expected size.

6. Worked example: metabolite overlap as a defined set operation

An overlap count is meaningful only after defining the eligible universe and key.

The current build found **153 shared nonblank exact RefMet strings before artifact and eligibility exclusions**. Eight [ST003 348](#) isotope-labeled or internal-standard rows mapped to ordinary RefMet labels; excluding those nonbiological rows yielded a **conservative biological overlap of 145**. Treat both numbers as versioned audit results: 153 tests the raw string-intersection step, while 145 tests the stated artifact filter. Neither is a shared-response count. The accepted overlap can change when:

- Internal standards, contaminants, or artifacts are excluded.
- Duplicate features mapping to the same RefMet name are collapsed.
- Annotation-resolution rules are applied.

- Review-required mappings are removed.
- Analysis modes are combined or kept separate.
- A repository or RefMet record is revised.

Required overlap report

Report a flow rather than one unexplained number:

1. Distinct submitted labels in each study.
2. Nonblank RefMet results.
3. Reviewed and accepted RefMet mappings.
4. Eligible mappings after artifact and feature-role rules.
5. Unique exact-name keys in each study.
6. Intersection size.
7. Diet-only and exercise-only sizes.
8. Duplicate-to-key counts.
9. Isotope/internal-standard exclusions, expected to reduce 153 to 145 in the audited build.

Also report the denominator used for percentages. [Intersection / diet set](#), [intersection / exercise set](#), and Jaccard similarity answer different questions.

What overlap does and does not mean

A shared standardized name shows that both studies reported an eligible feature under the selected key. It does not show:

- Equal concentration.
- Equal detectability or coverage.
- Equal identification confidence.
- Equal structural resolution.
- A shared intervention response.
- A pathway effect.

Absence from one study can reflect biological absence, platform coverage, annotation, filtering, ionization, detection limit, or reporting choices.

7. Class-level summaries

RefMet classifications can support a broad view of the reviewed shared-name set. Use counts or proportions of eligible unique names, not pooled peak areas. State how unclassified records and multi-mapped features are handled.

A defensible summary might say:

The 145-name conservative overlap has RefMet class annotations for all 145 names. Amino acids and peptides, fatty acids, and fatty esters are the three largest classes in this shared-name set. These counts describe nomenclature coverage of the overlap and should not be interpreted as class abundance in plasma or serum.

Accessible description of the class-summary figure

One horizontal bar chart shows the 12 most frequent RefMet main classes among the 145 conservative shared names. The leading bars are amino acids and peptides (37), fatty acids (25), fatty esters (15), purines (11), bile acids (8), and pyridine alkaloids (7); the remaining six displayed classes have four or fewer names each. The x-axis is [Conservative shared RefMet names](#). Bars are counts of standardized names, not metabolite concentrations or biological class abundance, and 24 names in less frequent classes are not shown in this top-12 view.

8. Principal component analysis: within each study

PCA summarizes directions of variation in a numeric matrix. It is sensitive to scale, skew, missingness, outliers, preprocessing, and feature selection. The notebook should create separate PCA models for each study or compatible analysis block.

The notebook carries out the implementation for you. You are not expected to understand every line of code. Focus on which samples and features enter each PCA, how missing and large values are handled, and whether the interpretation stays within the study design.

Pre-PCA audit

For each PCA, record:

- Included study, analysis modes, samples, and features.
- Biological versus QC sample roles.
- Missingness threshold and handling.
- Zero and near-zero variance removal.
- Transformation, centering, and scaling.
- Treatment of duplicate RefMet mappings.
- Explained variance for displayed components.

Use transformations only when compatible with the measurement scale and documented. Do not add an arbitrary pseudocount without recording its value and evaluating sensitivity. Fit imputation and scaling rules within the analyzed study; a combined fit across studies can transmit study-specific scale information.

Diet-study PCA

Color or facet the biological samples by diet and shape them by timepoint, while keeping the concurrent antibiotic/PEG intervention visible in the caption or an annotated time axis. Display pooled QC samples separately if they are used diagnostically. A visible time pattern is not automatically a diet effect.

Questions to ask:

- Do pooled QC samples cluster more tightly than biological samples?
- Are extreme points associated with missingness or a specific analytical mode?
- Is variation aligned with time, diet, sex, participant, or batch metadata?
- Does the pattern change after reasonable sensitivity checks?
- Can diet be separated from setting and assignment differences for all groups? Not without additional assumptions.

Exercise-study PCA

Color samples by [REST](#), [STAT](#), [REC3](#), and [REC22](#), and connect repeated observations from the same athlete only after participant linkage is validated. The four labels describe an acute and recovery trajectory; they are not interchangeable with diet-study days.

Questions to ask:

- Is any separation ordered along the known recovery sequence?
- Are trajectories consistent across athletes or driven by a subset?
- Do positive- and negative-ion analysis blocks behave differently?
- Could collection clock time, fasting status, or processing contribute?

Why this module omits a combined PCA

Study identity is perfectly associated with diet-versus-exercise context and also differs in matrix, platform implementation, units, laboratory, population, and protocol. A PCA of the stacked raw matrices would primarily encode those inseparable study-of-origin differences. This module therefore omits that plot and teaches the confounding problem through the metadata audit plus two independent within-study PCA figures whose numeric axes are not compared.

Accessible descriptions of the PCA figures

The diet figure is titled [ST001521 / AN002534: within-study PCA](#) and contains 150 biological plasma samples; the 10 pooled QPP samples are excluded. Color represents Western, Vegan, or Modulen, and marker shape represents Baseline, Day 5, Day 9, Day 12, or Day 15. PC1 explains 34.4% and PC2 explains 11.6% of variance.

The exercise figure is titled [ST003 348 / AN005 483: within-study PCA](#) and contains 76 serum samples. Color represents Rest, Immediate post, 3 h recovery, or 22 h recovery; the plot does not draw participant-trajectory lines. PC1 explains 16.1% and PC2 explains 8.4% of variance. The two PCA models have independent coordinate systems, so point distances and axis values must not be compared between figures.

9. Hands-on notebook activity

Part A - Retrieval audit

1. Review [NB-L1](#) and run the [NB-L2](#) retrieval cells if they have not already been executed in this fresh kernel.
2. Confirm both locked accessions in the output.
3. Record live or cached status for each endpoint.
4. Compare the structural audit with the expected checks above.
5. Investigate any discrepancy before continuing.

Part B - Sample-role and time audit

1. In [NB-L2](#), identify the 10 [QPP...](#) rows in [ST001 521](#) and verify that they are not counted as participants.
2. Create explicit physiological labels for the exercise study: pre-exercise, immediate post-exercise, 3-hour recovery, and 22-hour recovery.
3. Preserve the diet-study days and annotate the antibiotic/PEG interval rather than mapping them to the exercise labels.
4. Record plasma versus serum as a partial compatibility decision.

Part C - Crosswalk and overlap

1. In [NB-L3-CROSSWALK](#), join the reviewed crosswalk using accession, analysis, and source-feature keys.
2. Inspect the join-cardinality report.
3. Reproduce the intermediate 153 shared nonblank exact RefMet strings if the same retrieval and preliminary rules are in use.
4. Apply the nonbiological artifact rule and verify that eight [ST003 348](#) isotope/internal-standard rows are excluded, producing 145 conservative biological overlaps in the audited build.
5. Apply any additional mapping-status and annotation-resolution eligibility rules and record every exclusion count.
6. In [NB-L4-CLASS](#), produce the set summary and class-count figure.

If 153 and then 145 are not reproduced, do not tune filters merely to make them appear. Check versions, retrieved analyses, missing-value definitions, isotope/internal-standard rules, duplicate collapse, and crosswalk state; document the reason for any legitimate difference.

Part D - PCA and sensitivity

1. Run [NB-L4-PCA-DIET](#) and [NB-L4-PCA-EXERCISE](#) separately.
2. Confirm QC samples are excluded from biological inference but available for assay diagnostics.
3. State the preprocessing choices in plain language.
4. Repeat one reasonable sensitivity analysis, such as changing the missingness threshold or analyzing ion modes separately.
5. Compare which observations persist and which change.

Part E - Write a bounded interpretation

Use this four-sentence scaffold:

1. **Observation:** "Within [ST...](#), the exploratory output shows..."
2. **Context:** "The relevant design and phenotype metadata indicate..."
3. **Alternative explanations:** "This pattern could also reflect..."
4. **Boundary:** "These data do not establish..."

Avoid the verbs [proves](#), [causes](#), [confirms](#), or [demonstrates](#) unless the design and analysis truly support them.

10. Optional extension: MetENP

MetENP can be introduced as an optional pathway and enrichment resource supported through CFDE. Export only reviewed identifiers in a format the tool documents, retain the background universe, and record tool and database versions. Pathway output inherits all upstream identification and coverage limitations. It is hypothesis-generating, not validation of a diet-versus-exercise mechanism.

11. Interpretation challenge

For each statement, mark [supported](#), [partially supported](#), or [unsupported](#), then revise it.

1. "The studies share 153 biologically identical metabolites."
2. "The diet study contains 160 participants."
3. "The exercise samples are blood, so they are the same matrix as diet plasma."
4. "PCA separates exercise timepoints, proving the race walk caused every measured change."
5. "An apparent class difference may reflect assay coverage as well as biology."

Suggested revisions are in the answer key.

12. Analysis boundaries

- Exclude pooled QC samples from biological sample counts and participant-level inference.
- Preserve plasma versus serum; do not call both simply blood in the analysis matrix.
- Diet-study time includes microbiota-depletion procedures; it is not a pure duration-of-diet axis.
- Analyze repeated measures with participant structure; do not treat 150 or 76 rows as independent people.
- A raw exact-name overlap is an intermediate audit value until artifacts, ambiguity, and duplicates are addressed.
- PCA is exploratory and does not assign causes to components.
- Do not compare coordinates or loadings from separately fit PCAs as if they shared a coordinate system.
- Do not quantitatively pool the two studies in this module.
- Report sensitivity to preprocessing and missingness choices.
- Phrase class summaries as coverage of reported eligible names, not class concentration.

13. Knowledge check

KC4-01. Why should the [QPP...](#) rows in [ST001521](#) not be included in biological PCA?

- A. They are pooled quality-control samples rather than participant-timepoint observations.
- B. PCA cannot use plasma.
- C. Their identifiers contain letters.
- D. Quality-control samples must always be deleted from the repository.

KC4-02. What does the intermediate overlap of 153 represent?

- A. 153 metabolites proven to respond to both diet and exercise.
- B. 153 shared nonblank exact RefMet strings before artifact and eligibility exclusions under a specific build.
- C. 153 directly comparable concentrations.
- D. A permanent repository constant.

KC4-03. Which PCA interpretation is defensible?

- A. A separate within-study PCA is an exploratory view whose pattern must be interpreted with preprocessing, repeated measures, and design context.
- B. Separation between the two studies proves a diet-versus-exercise mechanism.
- C. Scaling removes plasma-versus-serum differences.
- D. PCA automatically corrects batch effects.

What the analysis can support

Validate each step before interpreting the result. A count can be reproducible and still support only a narrow claim, so use the study metadata to state that claim clearly.

Primary sources and first-party documentation

1. Metabolomics Workbench. [ST001521 study record](#) and [REST summary](#). Accessed August 10, 2026.
2. Metabolomics Workbench. [ST003348 study record](#) and [REST summary](#). Accessed August 10, 2026.
3. Metabolomics Workbench. [REST Service, version 1.2](#). Updated July 22, 2025; accessed August 10, 2026.
4. Metabolomics Workbench. [RefMet database](#). Accessed August 10, 2026.
5. Jolliffe IT, Cadima J. [Principal component analysis: a review and recent developments](#). *Philosophical Transactions of the Royal Society A*. 2016;374:20150202.
6. Metabolomics Workbench. [MetENP demonstration and documentation](#). Accessed August 10, 2026.

Lesson 5 - Access patterns and transfer to additional resources

Estimated time: 15 minutes

Bloom levels: Evaluate, create

Module objective addressed: LO5

Learning objectives

By the end of this lesson, you will be able to:

1. Determine the current access pattern of a specific dataset and intended action using dated first-party evidence.
2. Identify which parts of the Metabo-Diet workflow transfer unchanged and which must be redesigned.
3. Draft a compliant analysis architecture for open, agreement-gated, or controlled-compute data.
4. Produce a go, revise, wait, or stop decision with unresolved requirements stated explicitly.

Lesson map

Activity	Minutes
Revisit access as a dataset property	3
Worked transfer cases	4
Transfer checklist	5
Knowledge check	3
Instructional subtotal, excluding posttest	15
Posttest, additional assessment time	8
Learner time with posttest	23

In the notebook (NB-L5): Review the source log and edit the transfer-decision cell with dated evidence for one exact target. Choose GO, REVISE, WAIT, or STOP, then run NB-REPRO. Complete the posttest after the artifact manifest appears without errors.

1. Access tier is not a logo

In Lesson 1, access was introduced as a planning problem. After completing an open-data workflow, you can now see exactly where access rules affect the analysis.

Access is determined by the combination of:

- A specific dataset and release.
- The requested data level.
- Your identity, institution, training, and approvals.
- The intended use.
- The compute location.
- What inputs and outputs may cross the environment boundary.
- Current terms and policies.

Do not assign a permanent tier to an entire program. Current MoTrPAC documentation, for example, distinguishes public releases that community users can access without an account from restricted or embargoed data that require authentication, a data-use agreement, or other authorization. The access pattern is dataset-dependent. All of Us provides Registered and Controlled Tier collections in a governed Researcher Workbench and restricts external release of participant-level data and small counts under its policies.

The July 2026 proposal used MoTrPAC as a simplified registration-gated example. This module intentionally updates that framing to match current first-party documentation: **verify the exact release and action every time.**

2. What stays the same, and what must change

Some harmonization practices transfer across environments:

- Define the scientific question and estimand.
- Preserve original identifiers and metadata.
- Record provenance and decision rules.
- Separate naming, specimen, time, and unit compatibility.
- Flag unresolved mappings.
- Validate join cardinality.
- Respect repeated measures.
- Bound interpretation by design.

Other parts may need redesign:

- Authentication and API calls.
- Where code executes.
- Where raw, derived, and cached data are stored.

- Whether external reference resources can be queried from the environment.
- Whether data can be joined across resources.
- Whether row-level outputs, logs, plots, models, or counts can be exported.
- Who reviews an export and what disclosure thresholds apply.
- How collaborators receive access.

You can reuse the analysis steps even when the data must remain in a governed workspace.

3. Worked transfer case A: a MoTrPAC dataset

Scenario

You find a MoTrPAC metabolomics release that may complement the exercise case study.

Verification sequence

1. Identify the exact collection, study, release, assay level, and files.
2. Open current MoTrPAC Data Hub and Knowledge Center documentation.
3. Test whether the desired public file can be browsed or downloaded without an account.
4. If the file is restricted or embargoed, document authentication, DUA, publication, confidentiality, and review requirements.
5. Record the evidence URL, page version if available, date, and result.

Architecture decision

- **Public file:** The local or Colab-style workflow may remain possible, subject to license, size, and terms. Replace the MW-specific retrieval adapter while retaining provenance and crosswalk logic.
- **Restricted file:** Obtain authorization before access. Store and analyze only in permitted locations; do not add restricted content to the module cache. Adapt collaboration and output review to the DUA.
- **Unclear status:** Pause data retrieval and ask the program help desk. Continue with public metadata or synthetic schema only.

Scientific decision

MoTrPAC's detailed exercise design can improve phenotype context, but repository membership does not make its measurements quantitatively compatible with [ST003348](#). Repeat the full platform, specimen, time, and unit assessment.

4. Worked transfer case B: Nutrition for Precision Health in All of Us

Scenario

You want to adapt the diet harmonization workflow to NPH data stored in the All of Us Researcher Workbench.

Verification sequence

1. Confirm that the desired NPH data type and release are available.
2. Determine whether the required collection is Registered or Controlled Tier.
3. Verify current institutional, researcher, identity, training, workspace, and project requirements.
4. Review the Data User Code of Conduct and Data and Statistics Dissemination Policy.
5. Determine what participant-level data, code outputs, counts, plots, models, or aggregate results may leave the Workbench.

Architecture decision

The safe default is to move approved code and public reference mappings **into** the governed workspace, run participant-level work there, and export only outputs permitted under current policy. Do not download NPH participant-level data to populate this module's local files or cached fallback. If external RefMet calls are unavailable or inappropriate, import a dated, approved reference table without participant data and record its version.

Scientific decision

NPH's modular diet design and multimodal context may support questions that the small FARM case cannot. It does not remove the need to define diet period, test-meal timing, specimen, assay, repeated measures, and the intended estimand.

5. Accessible description of the transfer diagram

The diagram has a central box labeled [portable harmonization logic](#). Three arrows point to: an open laptop, a sign-in workspace, and a locked cloud workspace. The logic moves to all three. Raw data arrows move freely only into the open laptop; gated data remain in approved storage; controlled participant-level data remain inside the locked cloud. A single outbound arrow from the cloud is labeled [policy-permitted reviewed outputs](#).

6. Hands-on activity: make a transfer decision

Use [module/templates/access_tier_transfer_checklist_learner.md](#) for a resource of your choice. You may use a specific MoTrPAC release, an NPH/All of Us collection, another Metabolomics Workbench accession, MetaboLights, or a consortium repository relevant to your work.

For the timed course, complete five items: the exact dataset and intended action, one first-party source with its check date, the access pattern, the compute/storage boundary, and a [GO](#), [REVISE](#), [WAIT](#), or [STOP](#) decision. The full checklist is a project extension.

Required evidence

- Exact resource, dataset, and version or release.
- Intended data and action.
- First-party access documentation URL and check date.
- Account, agreement, institutional, training, or approval requirements.
- Permitted compute and storage location.
- Input and output movement rules.
- Metadata and identifier availability.
- Required changes to retrieval, cache, crosswalk, and reporting.
- One scientific compatibility risk distinct from access.

Decision options

- [GO](#): Requirements are verified and the planned workflow is permitted and scientifically appropriate.
- [REVISE](#): The goal is possible, but architecture, scope, output, or analysis must change.
- [WAIT](#): A documented approval, release, or clarification is pending.
- [STOP](#): The planned action is prohibited or cannot answer the scientific question.

Write one sentence that distinguishes policy evidence from your interpretation. Example:

The first-party page states that this public release can be downloaded without an account; I infer that the local retrieval step is feasible, but I still need to confirm license and scientific compatibility before reuse.

7. Safety and governance rules

- Verify current first-party terms; do not rely on this lesson as legal or policy authority.
- Dataset landing-page visibility does not prove participant-level download permission.
- Public metadata can be useful even when data are restricted, but it does not authorize reconstruction or access.
- Do not place credentials, tokens, signed URLs, or controlled data in notebooks, caches, logs, version control, or training artifacts.

- Do not evade a governed environment by exporting row-level data, small cells, recoverable embeddings, or model artifacts that policy treats as sensitive.
- Access approval does not establish scientific comparability.
- If rules conflict or remain unclear, pause the affected data movement and seek the resource's official support channel.

8. Knowledge check

KC5-01. What is the most accurate access description for MoTrPAC?

- A. Every MoTrPAC dataset requires registration.
- B. Every MoTrPAC dataset is fully open.
- C. Access is dataset- and release-dependent; public and restricted paths exist and must be verified.
- D. MoTrPAC data can be used only inside All of Us.

KC5-02. What should move into a controlled-compute environment?

- A. Approved analysis code and public reference resources that comply with environment policy.
- B. A public link that silently downloads participant data to a local laptop.
- C. Shared credentials.
- D. A module cache populated with participant-level records.

KC5-03. A dataset is publicly downloadable and scientifically incompatible with your target estimand. What is the correct decision?

- A. Pool it because access is open.
- B. Treat access and scientific compatibility separately; revise or stop the integration.
- C. Standardize names and ignore the design.
- D. Call it controlled access.

9. Posttest and transfer commitment

Complete [module/assessments/posttest.json](#) through the delivery platform, then finish this statement:

Within the next month, I will apply the workflow to _____. My first evidence check will be _____, and the first incompatibility I will test is _____.

Final decision rule

The harmonization workflow transfers when its logic, provenance, and guardrails are preserved. Access changes the analysis architecture; study design changes the claims. Both must be checked before data move or models run.

Primary sources and first-party documentation

1. MoTrPAC. [Data access FAQ](#). Accessed August 17, 2026.
2. MoTrPAC. [Data Hub access documentation](#) and [consortium/restricted onboarding](#). Accessed August 17, 2026.
3. NIH Common Fund. [Nutrition for Precision Health FAQ](#). Accessed August 10, 2026.
4. All of Us Research Program. [Researcher Workbench overview](#). Accessed August 10, 2026.
5. All of Us Research Program. [Policy questions and dissemination guidance](#). Accessed August 10, 2026.
6. All of Us Research Program. [Research project directory and access-tier requirements](#). Accessed August 10, 2026.

Learner posttest

Complete after all five lessons and the notebook. The delivery platform should hide answers and rationales until submission.

Estimated time: 8 minutes

1. The ST001521 factors response contains 160 rows. Which interpretation is supported by the module audit?

- ☐ A. There are 160 independent participants.
- ☐ B. There are 150 participant-timepoint rows and 10 pooled QPP quality-control rows in the current record.
- ☐ C. There are 40 participants measured four times.
- ☐ D. The 10 QPP rows should be treated as a fourth diet group.

Response: _____ Confidence (1-5): _____

2. How should the ST001521 plasma and ST003348 serum specimens be handled?

- ☐ A. Rename both blood and pool their peak areas.
- ☐ B. Keep exact matrix labels, optionally use a broader blood-derived-fluid category for discovery, and prohibit matrix-equivalent quantitative pooling.
- ☐ C. Discard both studies.
- ☐ D. Convert serum to plasma using z-scores.

Response: _____ Confidence (1-5): _____

3. Why should ST001521 Day 9 not be mapped to ST003348 REC3 as a generic post timepoint?

- ☐ A. Day 9 is not numeric.
- ☐ B. Day 9 belongs to a feeding and antibiotic/PEG protocol, while REC3 is 3 hours after an endurance exercise bout.
- ☐ C. REC3 is a quality-control sample.
- ☐ D. Metabolomics Workbench prohibits time mapping.

Response: _____ Confidence (1-5): _____

4. Which statement best describes a direct cross-study diet-versus-exercise quantitative contrast?

- ☐ A. It isolates phenotype because both studies use LC-MS.
- ☐ B. It is confounded by study, population, matrix, laboratory, platform implementation, time, and protocol, so the module does not use it for causal inference.
- ☐ C. It becomes causal after RefMet mapping.
- ☐ D. It is valid whenever sample sizes exceed 20.

Response: _____ Confidence (1-5): _____

5. ST003348 contains isotope/internal-standard labels that map to ordinary RefMet names. What should the biological overlap workflow do?

- ☐ A. Count them as endogenous overlaps because RefMet is nonblank.
- ☐ B. Preserve the source labels, classify their nonbiological analytical role, and exclude them from biological overlap with a reason.
- ☐ C. Delete them without a record.
- ☐ D. Use them to calibrate ST001521 automatically.

Response: _____ Confidence (1-5): _____

6. Two source features in one analysis map to the same RefMet name. What is the reproducible approach?

- ☐ A. Average their intensities immediately and discard source IDs.
- ☐ B. Keep feature-level rows and provenance, flag the many-to-one mapping, and define a separate collapse rule for any set-level summary.
- ☐ C. Choose the feature with the larger mean.
- ☐ D. Count the RefMet name twice in the overlap set.

Response: _____ Confidence (1-5): _____

7. Which overlap report matches the audited module build?

- ☐ A. 153 proven shared biological responses
- ☐ B. 153 shared nonblank exact RefMet strings before exclusions; 145 in the conservative biological set after excluding eight ST003348 isotope/internal-standard rows, with all rules versioned
- ☐ C. 145 directly comparable concentrations
- ☐ D. The overlap is undefined because no names can ever be compared

Response: _____ Confidence (1-5): _____

8. Which PCA plan follows the module guardrails?

- ☐ **A.** Fit separate, documented within-study PCAs; keep repeated measures and sample roles visible; do not stack the uncalibrated raw matrices.
- ☐ **B.** Pool both studies and label the first component diet versus exercise.
- ☐ **C.** Include pooled QPP samples as participants.
- ☐ **D.** Hide preprocessing because PCA is objective.

Response: _____ Confidence (1-5): _____

9. The live REST request fails and the notebook uses a cached response. What must the analysis report?

- ☐ **A.** Nothing; cache use is invisible to learners.
- ☐ **B.** The failed endpoint/status, cached retrieval time, integrity check, and a visible cached-source flag.
- ☐ **C.** Only the final figure.
- ☐ **D.** A fabricated current access time.

Response: _____ Confidence (1-5): _____

10. How should MoTrPAC access be described when transferring this workflow?

- ☐ **A.** Uniformly registration-gated
- ☐ **B.** Uniformly unrestricted
- ☐ **C.** Dataset- and release-dependent, with public and restricted paths that require a current first-party check
- ☐ **D.** Available only through the Metabolomics Workbench REST API

Response: _____ Confidence (1-5): _____

11. What is the appropriate default plan for participant-level NPH data in All of Us?

- ☐ **A.** Copy data to the module's public cache.
- ☐ **B.** Run approved code in the Researcher Workbench, keep governed data there, and export only policy-permitted outputs after checking current requirements.
- ☐ **C.** Share a colleague's credentials.
- ☐ **D.** Assume the MW license applies.

Response: _____ Confidence (1-5): _____

12. A crosswalk join unexpectedly doubles the number of measurement rows. What should happen next?

- ☐ **A.** Proceed because more rows improve power.
- ☐ **B.** Drop duplicated rows at random.
- ☐ **C.** Stop, inspect key uniqueness and join cardinality, and revise the documented join rule before analysis.
- ☐ **D.** Average all values across both studies.

Response: _____ Confidence (1-5): _____

Cohort-comparison worksheet - learner version

Module: Metabo-Diet

Learner: _____

Date: _____

Retrieval timezone: _____

Purpose

Use this worksheet to compare [DIET_ACCESSION = ST001 521](#) and [EXERCISE_ACCESSION = ST003 348](#) without assuming that a shared label makes records quantitatively exchangeable.

Use these codes:

- [D](#) - directly comparable for the stated purpose
- [P](#) - partially comparable; retain the difference and information loss
- [N](#) - not comparable for the stated purpose
- [U](#) - not yet assessable from available evidence
- [NR](#) - searched but not reported
- [NA](#) - not applicable

Every factual entry needs a source URL, mwTab block, REST endpoint, publication location, or explicit missingness code.

1. Comparison purpose and estimand sketch

Choose one primary purpose:

- ☐ Metadata inventory
- ☐ Exact standardized-name overlap
- ☐ Broad RefMet class coverage
- ☐ Within-study change followed by qualitative pattern comparison
- ☐ Other non-pooled purpose: _____

Population: _____

Exposure/intervention: _____

Outcome or analyte set: _____

Biological time definition: _____

Summary measure: _____

What you will not use this comparison to claim: _____

For the timed lesson, complete the eight named rows in Section 9 plus the supporting fields in Sections 2, 3, 5, and 6. The remaining rows are a project extension.

2. Dataset identity and access evidence

Field	Diet study	Exercise study
Configuration key	DIET_ACCESSION	EXERCISE_ACCESSION
Resolved accession		
Study title		
Project/study DOI		
Study landing page		
REST summary endpoint		
License shown		
Public status check result		
Check timestamp, UTC		
Live or cached retrieval		
Cache timestamp/checksum, if used		

3. Study design and population

Field	Diet study evidence	Exercise study evidence	D/P/N/U	Justification and intended use
Species				
Study design				
Recruitment/setting				
Inclusion/exclusion features				
Participants recruited				
Participants analyzed				
Biological samples				

Field	Diet study evidence	Exercise study evidence	D/P/N/U	Justification and intended use
QC/blank/other samples				
Unit of assignment				
Unit of analysis				
Repeated-measures linkage				
Randomization				
Blinding/masking				
Attrition				

4. Phenotype anchor and intervention

Field	Diet study evidence	Exercise study evidence	D/P/N/U	Justification and intended use
Phenotype anchor				
Intervention/exposure groups				
Dose or intensity				
Duration				
Adherence or supervision				
Feeding state				
Concurrent interventions				
Behavioral/clinical covariates				
Assignment/selection caveat				

5. Specimen and pre-analytical context

Field	Diet study evidence	Exercise study evidence	D/P/N/U	Justification and intended use
Broad factor label				
Specific biological matrix				
Collection method/tube				
Processing				

Field	Diet study evidence	Exercise study evidence	D/P/N/U	Justification and intended use
Storage				
Collection clock time				
Fasting status/duration				
Pre-analytical gaps				

6. Timepoint semantics

Do not use the same harmonized time label merely because both studies contain [baseline](#), [rest](#), or [post](#) concepts.

Diet study timepoints

Source label	Anchor event	Offset/unit	Physiological/protocol state	Visit/order	Source	Resolved?

Exercise study timepoints

Source label	Anchor event	Offset/unit	Physiological/protocol state	Visit/order	Source	Resolved?

Time-axis compatibility decision: _____

Evidence-based reason: _____

7. Assay and measurement context

List one row per analysis ID.

Accession	Analysis ID	Analysis summary	Chromatography	Instrument	Ion mode	Targeted status	Units/scale	Source

Can quantitative values be pooled across studies? Yes / No / Not yet assessable

Reason: _____

8. Metadata provenance conflicts

Record fields for which two source locations differ or one source is more specific.

Accession	Field	Source A/value	Source B/value	Resolution rule	Preserved evidence

9. Comparability decision matrix

Complete at least eight rows. Name the purpose in each row; a field can receive a different decision for a different purpose.

Domain/field	Intended purpose	D/P/N/U	Evidence-based justification	Allowed action	Prohibited inference
Species					
Matrix					
Timepoint					
Participant count					
Repeated-measures structure					
Phenotype anchor					
RefMet name					

Domain/field	Intended purpose	D/P/N/U	Evidence-based justification	Allowed action	Prohibited inference
Annotation resolution					
Units/scale					
Platform/mode					
Other: _____					

10. Decision summary

Directly comparable elements: _____

Partially comparable elements: _____

Elements that must remain separate: _____

Unresolved evidence: _____

Complete the required exclusion statement:

We will not combine _____ across the two studies because _____.
 We can still compare _____ by _____.

Write one bounded comparison claim:

Write one tempting but unsupported claim you will avoid:

Completion check

- ☐ Every factual entry has a source or explicit **NR**, **NA**, or **U** code.
- ☐ Participant counts are separate from biological sample and QC counts.
- ☐ Plasma and serum remain visible.
- ☐ All time labels have protocol meanings or are unresolved.
- ☐ Concurrent interventions and selection/assignment caveats are recorded.
- ☐ All analysis IDs and units are listed.
- ☐ At least one field is intentionally not harmonized.
- ☐ No conclusion relies on pooled cross-study quantitative values.

Metabolite and metadata crosswalk - learner version

Module: Metabo-Diet

Learner/reviewer: _____

Crosswalk version: _____

Created: _____

RefMet retrieval date/version: _____

What this worksheet is for

This template records source evidence, harmonized representations, review decisions, and purpose-specific eligibility for `DIET_ACCESSION = ST001 521` and `EXERCISE_ACCESSION = ST003 348`.

It supports auditable name and metadata comparison. It does not authorize pooling quantitative values across the studies.

Controlled values

Mapping status

- `accepted_exact`
- `accepted_broader`
- `review_required`
- `unmapped`
- `excluded_nonbiological`
- `not_evaluated`

Annotation resolution

- `discrete_structure`
- `stereochemistry_unspecified`
- `lipid_molecular_species`
- `lipid_sum_composition`
- `compound_class_only`
- `unknown`

Eligibility

Use `yes`, `no`, or `review` separately for:

- `eligible_exact_name_overlap`

- `eligible_class_summary`
- `eligible_within_study_quantitative`
- `eligible_cross_study_quantitative`

For this module, `eligible_cross_study_quantitative` is `no` for both studies.

Tab A - Metabolite mapping

Use one row per source feature per analysis. Do not deduplicate before preserving source identity.

Column	Required meaning
<code>row_id</code>	Stable crosswalk-row identifier
<code>config_key</code>	<code>DIET_ACCESSION</code> or <code>EXERCISE_ACCESSION</code>
<code>study_accession</code>	<code>ST001 521</code> or <code>ST003 348</code>
<code>analysis_id</code>	Source <code>AN...</code> identifier
<code>source_feature_id</code>	Original feature/metabolite ID if present
<code>source_metabolite_name</code>	Exact deposited label
<code>source_refmet_name</code>	RefMet value returned in the MW metabolite record, including blank
<code>source_formula</code>	Original formula or <code>NR</code>
<code>source_inchikey</code>	Original InChIKey or <code>NR</code>
<code>source_external_ids</code>	PubChem, HMDB, KEGG, or other IDs with namespaces
<code>source_unit</code>	Exact analysis unit or scale
<code>source_endpoint</code>	Exact URL or cached file identifier
<code>retrieved_at_utc</code>	ISO 8601 timestamp
<code>lookup_string</code>	Exact string sent to RefMet, if queried separately
<code>lookup_transform</code>	Changes from source label to lookup string; use <code>none</code> if unchanged
<code>refmet_candidate</code>	Candidate standardized name; preserve multiple candidates
<code>refmet_query_url</code>	Exact query or versioned bulk-table source
<code>annotation_resolution</code>	Controlled value above
<code>identification_evidence</code>	Evidence reported by source; do not upgrade it
<code>mapping_status</code>	Controlled value above
<code>decision_reason</code>	Evidence-based rationale
<code>reviewer_or_rule</code>	Person, rule ID, or software version

Column	Required meaning
reviewed_at_utc	ISO 8601 timestamp
eligible_exact_name_overlap	yes, no, or review
eligible_class_summary	yes, no, or review
eligible_within_study_quantitative	yes, no, or review
eligible_cross_study_quantitative	no for this module
exclusion_reason	Required when an eligibility field is no
decision_log	Append-only note for changes

Blank working rows

For the timed lesson, complete three rows: one retained shared name, one isotope-labeled or internal-standard exclusion, and one ambiguous or unmapped example. For a full project audit, complete at least 12 rows (six per study) and include a lipid when present.

row_id	config_key	analysis_id	source_metabolite_name	source_refmet_name	candidate	resolution	status	exact overlap	class summary	within-study	cross-study	decision reason
M001											no	
M002											no	
M003											no	
M004											no	
M005											no	
M006											no	
M007											no	
M008											no	
M009											no	
M010											no	
M011											no	
M012											no	

Attach or export the complete column set for machine use. The shortened table above is for review discussion only.

Tab B - Metadata-field mapping

Use one row per source field or coded value.

Column	Meaning
metadata_row_id	Stable identifier
config_key	Study configuration key
study_accession	Resolved accession
source_block_or_endpoint	mwTab block or REST URL
source_field	Exact field name
source_value	Exact value
construct_definition	Meaning supported by protocol
harmonized_field	Derived common field, if any
harmonized_value	Derived value, if any
transformation_rule	Reproducible derivation
information_lost	Detail removed or none
compatibility	Direct, partial, not comparable, or not yet assessable
purpose	Analysis purpose for that decision
source_url	Exact evidence location
retrieved_at_utc	Retrieval timestamp
reviewer_or_rule	Reviewer or rule version
decision_note	Rationale and restrictions

Required metadata rows

metadata_row_id	config_key	source field/value	construct	harmonized field/value	rule	information lost	compatibility	purpose	decision note
MD001			Specimen matrix						
MD002			Specimen matrix						
MD003			Timepoint						
MD004			Timepoint						
MD005			Timepoint						

metadata_row_id	config_key	source field/value	construct	harmonized field/value	rule	information lost	compatibility	purpose	decision note
MD006			Participant/sample relationship						
MD007			Phenotype						
MD008			Measurement scale						

Tab C - Unit and scale decisions

config_key	analysis_id	source quantity	source unit/scale	candidate target	dimensionally convertible?	quantification compatible?	action	formula/rule	reason

Tab D - Cross-study overlap audit

Do not report only the final intersection.

Stage	Diet count	Exercise count	Shared count	Rule/version	Exclusions at this stage
Distinct submitted labels			NA		
Nonblank source RefMet strings					
Reviewed accepted mappings					
Eligible after artifact rules					
Unique exact-name keys					
Final reported set					

Build checks: The current release reported 153 shared nonblank exact RefMet strings before artifact and eligibility exclusions. Excluding eight [ST003348](#) isotope/internal-standard rows that mapped to ordinary RefMet labels yielded a conservative biological overlap of 145. If you do not reproduce 153 and then 145 with the same source version and rules, investigate and document the difference; do not tune decisions to force either count.

Reproducibility audit

- [] Original labels and identifiers are never overwritten.

- [] Every row includes accession, analysis ID, and source endpoint.
- [] Lookup transformations and query strings are recorded.
- [] Automated mappings are distinguishable from reviewed decisions.
- [] Annotation resolution is no more specific than source evidence.
- [] One-to-many and many-to-one cases remain visible.
- [] Internal standards, isotope-labeled compounds, pooled QC, and artifacts have explicit roles.
- [] Unmapped and review-required rows remain in the crosswalk.
- [] Eligibility is purpose-specific.
- [] Cross-study quantitative eligibility is [no](#).
- [] Decision changes are append-only or versioned.

Access-pattern and transfer checklist - learner version

Learner: _____

Date: _____

Resource support contact, if known: _____

Instructions

Assess one exact dataset, release, and intended action. Use current first-party documentation. Do not infer access from the program name, a search-result snippet, or a past release.

This checklist supports planning and is not legal advice or a replacement for the resource's terms, institutional review, or data-governance process.

For the timed lesson, complete five items: the exact dataset and action in Section 1, one first-party source and date in Section 2, one planning pattern in Section 3, the compute/storage boundary in Section 6, and the decision in Section 8. Use the remaining rows for a full project transfer.

1. Scope the request

Item	Entry
Program/resource	
Exact dataset or collection	
Release/version/date	
Assay/data level	
Requested fields/files	
Intended scientific question	
Intended action: view/query/download/join/analyze/export/share/publish	
Intended users/collaborators	
Intended compute location	
Intended storage location	
Intended outputs	

2. Record first-party evidence

Evidence question	Answer	First-party URL/document section	Checked at UTC	Status: verified/conflicting/unclear
Is the exact release discoverable?				
Is it released, embargoed, or pending?				
Can it be viewed without an account?				
Can the desired data be downloaded?				
Is an individual account required?				
Is institutional authorization required?				
Is training or identity verification required?				
Is a DUA, code of conduct, or click-through agreement required?				
Is project or access-committee approval required?				
Must analysis occur in a governed workspace?				
Can participant-level data leave the environment?				
What aggregate/count/output rules apply?				
Can external reference data or APIs enter the environment?				
Can code leave the environment, and under what conditions?				
Can collaborators receive or share access?				
What citation/acknowledgment is required?				
What license or reuse terms apply?				

3. Assign a planning pattern

Select only after completing the evidence table.

- ☐ **Open public retrieval:** The intended public data action is available without personal credentials.

- ☐ **Account/agreement/approval gated:** The intended action requires identity, terms, a DUA, or approval.
- ☐ **Controlled compute:** Participant-level work must remain in a governed environment and only permitted outputs leave.
- ☐ **Mixed:** Different files, levels, actions, or releases use different patterns.
- ☐ **Unclear:** Current evidence is insufficient or conflicting.

Evidence statement:

The first-party source states _____.

Your interpretation:

For my intended action, I infer _____.

4. Check scientific fit separately

Check	Evidence	Compatible/partial/not compatible/unclear	Required action
Population and design			
Phenotype/exposure definition			
Specimen and pre-analytics			
Timepoint semantics			
Assay/platform and coverage			
Units/scale and preprocessing			
Identifier/annotation resolution			
Repeated-measures structure			
Missingness and QC metadata			

Scientific comparison the data can support: _____

Scientific comparison the data cannot support: _____

5. Adapt the Metabo-Diet workflow

Component	Keep unchanged	Modify	Remove/replace	Planned implementation
Question and estimand	☐	☐	☐	
Retrieval adapter/API	☐	☐	☐	

Component	Keep unchanged	Modify	Remove/replace	Planned implementation
Authentication handling	☐	☐	☐	
Local public cache	☐	☐	☐	
Provenance manifest	☐	☐	☐	
Specimen/time crosswalk	☐	☐	☐	
RefMet mapping	☐	☐	☐	
QC/sample-role logic	☐	☐	☐	
Within-study analysis	☐	☐	☐	
Cross-resource join	☐	☐	☐	
Output disclosure review	☐	☐	☐	
Collaboration/sharing	☐	☐	☐	
Publication/citation	☐	☐	☐	

6. Data-flow sketch

Write each boundary explicitly.

```
[Public documentation/reference] --permitted input?--> [Compute environment]
[Governed/raw data] -----stays where?-----> [Compute environment]
[Derived participant-level data] ---stays where?-----> [Approved storage]
[Aggregate/output] -----review rule?-----> [Permitted destination]
[Code] -----review rule?-----> [Repository/collaborator]
```

Credentials/tokens location: _____

Credential rule: Never enter credentials or signed URLs in this worksheet, a notebook, a cache, or version control.

7. Risk and dependency log

Risk or dependency	Evidence	Owner	Mitigation	Stop condition
Access approval				
Dataset release/change				
Compute/storage policy				
Export/disclosure				
Identifier mapping				

Risk or dependency	Evidence	Owner	Mitigation	Stop condition
Scientific compatibility				
Other				

8. Decision

- ☐ **GO** - requirements are verified and the planned action is permitted and scientifically appropriate.
- ☐ **REVISE** - the goal is possible, but architecture, scope, output, or analysis must change.
- ☐ **WAIT** - a documented approval, release, or clarification is pending.
- ☐ **STOP** - the planned action is prohibited or cannot answer the scientific question.

Decision statement: _____

Unresolved items: _____

Next evidence check and date: _____

Final safety check

- ☐ I evaluated an exact dataset, release, and action.
- ☐ I used current first-party evidence and recorded a date.
- ☐ I separated policy evidence from my inference.
- ☐ I did not copy credentials or governed data into the worksheet.
- ☐ I separated access feasibility from scientific comparability.
- ☐ I identified where data, code, and outputs may move.
- ☐ I identified an official support path for unclear requirements.
- ☐ My decision can be revisited when the release or policy changes.

Metabo-Diet glossary

Terms are defined for this module's workflow. A repository or program may use a term more narrowly; prefer the source's current data dictionary when applying the workflow elsewhere.

A

Access pattern

The practical combination of identity, agreement, approval, compute, storage, and output rules governing a specific dataset, release, and action. This module uses open, agreement- or approval-gated, and controlled-compute patterns as planning aids rather than universal legal categories.

Acute exercise response

A physiological or molecular change associated with a single exercise bout. Its interpretation depends on exercise modality, intensity, duration, specimen timing, feeding state, and recovery interval.

Adduct

An ion formed when a molecule associates with or loses a species during ionization, such as a proton or sodium ion. An adduct label is not automatically a unique metabolite identity.

Analysis identifier

A Metabolomics Workbench [AN...](#) identifier for a particular assay or analysis within a study. A study accession may contain multiple analyses with different chromatography, ion mode, platform, or data tables.

Annotation resolution

The structural detail supported for an analyte, such as a discrete structure, lipid molecular species, lipid sum composition, compound class, or unknown feature.

API endpoint

A specific URL that returns one type of data from a service. This module requests separate Metabolomics Workbench endpoints for the study summary, factors, analyses, metabolites, and measurement data.

B

Batch effect

Systematic variation associated with processing order, laboratory conditions, instrument behavior, reagent lot, or another technical grouping rather than the target biology.

Biological sample

A specimen collected from a participant or experimental unit for biological inference. It is distinct from a pooled QC, blank, or technical standard.

C**Cached fallback**

A versioned copy of a public API response used when live retrieval is unavailable. It must have a retrieval timestamp and integrity evidence and must be labeled as cached when used.

CFDE

Common Fund Data Ecosystem, an NIH Common Fund effort supporting discovery, access, interoperability, and reuse across Common Fund data resources.

Chemical class

A group of compounds sharing a defined chemical relationship. A class-level count of reported metabolites describes dataset coverage, not the total concentration or abundance of that class.

Clinical or behavioral phenotype

A measured characteristic or exposure such as diet pattern, exercise dose, age, disease status, fitness, or adherence. The protocol and timing define what the field means.

Code cell

A notebook block that contains executable code. In Jupyter, click the cell and press **Shift+Enter** to run it.

 means the cell is running and a number means it finished.

Controlled-compute environment

An approved workspace in which governed data remain and analyses run. Only policy-permitted outputs may leave the environment.

Cross-study pooling

Combining participant-level or sample-level quantitative values from separate studies into one analysis as though they formed a common measurement process. Name harmonization alone does not justify pooling.

Crosswalk

A provenance-preserving table linking source fields or analytes to harmonized representations, with transformation rules, evidence, information loss, status, and downstream eligibility.

D

Data provenance

Evidence describing where a value came from and how it was retrieved, transformed, reviewed, and used. In this module it includes accession, analysis ID, endpoint, timestamp, source label, mapping query, rule version, and decision.

Data-use agreement (DUA)

An agreement defining permitted use, security, confidentiality, sharing, publication, and other responsibilities for a dataset. Requirements are resource- and release-specific.

Directly comparable

Two fields represent the same construct at sufficient granularity for a stated use, with compatible coding or a transparent lossless conversion.

E

EEN

Exclusive enteral nutrition. In the [ST001521](#) study summary, an EEN diet is one of the diet conditions. The deposited factor label [Modulen](#) refers to the named EEN product in the protocol; preserve both the source label and protocol interpretation.

Eligibility rule

An explicit rule determining whether a sample, feature, or mapping participates in a particular analysis. Eligibility may differ for exact-name overlap, class summaries, QC diagnostics, and quantitative analysis.

Estimand

A precise description of the quantity a study or analysis aims to estimate, including population, exposure or intervention, outcome, time, and summary measure.

F

FAIR

Findable, accessible, interoperable, and reusable. FAIR does not mean unrestricted or free of governance; controlled data can be FAIR when access conditions are clear and machine-actionable.

FARMM

Food And Resulting Microbial Metabolites, the study context represented by [ST001521](#). Its longitudinal design includes three diet groups and an antibiotic/polyethylene glycol intervention.

Fasting state

Whether and for how long a participant abstained from caloric intake before collection. A binary [fasted](#) field and a measured fasting duration are only partially comparable for some purposes.

Feature

An analytical signal, often defined by mass-to-charge ratio, retention time, and other attributes. An untargeted feature may be unidentified or map ambiguously to compounds.

H**Harmonization**

The documented alignment of representations while preserving provenance, incompatibility, and information loss. It improves structural comparability; it does not guarantee statistical exchangeability.

HMDB ID

An identifier in the Human Metabolome Database. It can provide a useful cross-reference but is not interchangeable with a RefMet name or proof of an experimental identification.

I**Identification confidence**

The evidence supporting a metabolite assignment. A standardized name does not increase the experimental evidence originally reported.

InChIKey

A fixed-length hash derived from an InChI chemical representation. Its blocks encode different layers of structural and protonation information. Representation differences can yield different keys for related forms.

Information loss

Detail removed or collapsed during harmonization, such as reducing fasting hours to a binary flag or combining plasma and serum under a broad blood-derived-fluid category.

Ion mode

Positive or negative ionization mode in mass spectrometry. Different modes can detect different subsets of metabolites and can create analysis-specific duplicates.

J

JupyterLab

The browser-based interface used to read and run the Python notebook. A Jupyter notebook mixes instructions, code cells, and saved outputs in one file.

Join cardinality

The relationship between keys in two tables: one-to-one, one-to-many, many-to-one, or many-to-many. An unexpected many-to-many join can duplicate measurements and must be resolved before analysis.

K

Kernel

The Python process that runs notebook code and stores objects created by earlier cells. Restarting the kernel clears that memory, which is why the module asks learners to run cells from top to bottom.

KEGG compound ID

An identifier used in the KEGG compound resource and pathway ecosystem. It is useful for pathway linking but is not a universal identity key.

L

LC-MS

Liquid chromatography coupled with mass spectrometry. Chromatography, ionization, instrument, acquisition, and preprocessing choices influence coverage and measurement values.

Longitudinal design

A design with repeated observations over time. Samples from the same participant are correlated and must not be treated as independent people.

M

Mapping status

The reviewed state of a crosswalk row, such as accepted exact, accepted broader, review required, unmapped, excluded nonbiological, or not evaluated.

Matrix

The biological material measured, such as plasma or serum. Plasma and serum are both blood-derived but differ in collection and composition and remain distinct in this module.

Metabolite overlap

The intersection of two explicitly defined analyte sets. The result depends on mapping status, annotation resolution, artifact exclusions, duplicate collapse, and denominator.

Metabolomics Workbench

The platform hosting the National Metabolomics Data Repository and tools including the REST service and RefMet.

MetENP

A CFDE-supported resource for metabolite enrichment and network/pathway interpretation. Its output inherits the identification, background-universe, and coverage limits of its inputs.

MoTrPAC

Molecular Transducers of Physical Activity Consortium. It develops molecular maps of physical activity and provides public and restricted data pathways depending on the specific release.

mwTab

The sectioned data and metadata format used by Metabolomics Workbench. Blocks can describe study, subject, factors, collection, sample preparation, analytical methods, and metabolite measurements.

N**NMDR**

National Metabolomics Data Repository, hosted through Metabolomics Workbench.

NPH

Nutrition for Precision Health, powered by the All of Us Research Program. Public NIH documentation describes a modular precision nutrition study whose data are stored and analyzed in the All of Us Researcher Workbench.

Not yet assessable

A compatibility status used when essential evidence is missing or unresolved. It must not be silently converted to comparable or not comparable.

Notebook

An [.ipynb](#) file containing instructions, code, and outputs. The primary notebook for this module is [metabo_diet_harmonization.ipynb](#).

P

Partially comparable

Two fields share a broader construct but differ in granularity, timing, protocol, matrix, or measurement. They may support a coarser or stratified use if the difference remains visible.

PCA

Principal component analysis, an unsupervised linear method that summarizes major directions of variance. PCA is sensitive to preprocessing and does not identify causal mechanisms.

Peak area

An integrated analytical signal. Peak areas from different assays are not made directly comparable merely by scaling or using the same metabolite name.

Phenotype anchor

The exposure or intervention context around which a study is organized, such as controlled feeding or an acute endurance exercise bout.

Physiological timepoint

A time label defined by an anchor event, offset, unit, and biological state, rather than only a code such as [post](#) or [T2](#).

Pooled QC

A quality-control sample created by combining aliquots, often used to monitor analytical stability. In [ST001521](#), [QPP...](#) rows are pooled QC candidates and are excluded from biological participant counts.

PubChem CID

A record identifier in PubChem. Record form and structural specificity must be reviewed when using it as a cross-reference.

R

REC3 and REC22

In [ST003348](#), collection labels for 3 hours and 22 hours after exercise, respectively.

RefMet

A reference nomenclature for metabolomics designed to standardize names for discrete structures and analytically reported metabolite species. A RefMet mapping is a naming result, not proof of identification or quantitative equivalence.

REST

Representational State Transfer. In this module, [REST](#) can also appear as the [ST003348](#) pre-exercise timepoint label. Context distinguishes the MW REST API from the exercise collection label.

REST API

Metabolomics Workbench's URL-based service for programmatic access to study, compound, RefMet, and other contexts.

S

Sample role

The analytical purpose of a row, such as biological, pooled QC, blank, technical standard, or unknown.

Serum

The liquid fraction obtained after blood clotting. The [ST003348](#) factors endpoint says [blood](#), while its collection metadata specifies serum; the module preserves both and uses serum as the supported specific matrix.

Specimen context

Matrix, anatomical source, collection conditions, processing, storage, fasting state, and related evidence needed to interpret a measurement.

STAT

In [ST003348](#), the immediately post-exercise collection label. It does not mean a generic post-intervention timepoint.

Structural harmonization

Alignment of names, fields, categories, and relationships. It does not by itself justify combining quantitative observations.

Study accession

A Metabolomics Workbench [ST...](#) identifier. The locked case studies are [ST001521](#) and [ST003348](#).

T

Targeted metabolomics

Measurement focused on a defined analyte panel, often with method-specific quantitative or semiquantitative procedures. It differs from untargeted coverage and must not be assumed to share scale or detection properties.

Time semantics

The biological meaning of a time label, including anchor event, offset, unit, state, visit or period, and repeated-measures order.

Terminal

A text-based application used to enter commands. On macOS it is commonly called Terminal; on Windows this module uses PowerShell.

Traceback

The red error report shown when a Python cell fails. Keep the message and stop at the failed cell; it identifies where to begin troubleshooting.

U

Unit reconciliation

A documented conversion performed only when the same quantity, compatible matrix and denominator, quantification basis, and conversion relationship are established.

Untargeted metabolomics

Broad profiling of analytical features without limiting measurement to a small predefined panel. Feature annotation and missingness require careful interpretation.

V

Virtual environment

A project-specific Python installation directory, named `.venv` in this module. It keeps the course packages separate from other Python projects and lets Jupyter use the tested versions.

W

Within-study analysis

An analysis fit separately inside one study using its own design, repeated measures, scale, and preprocessing. It is the default quantitative strategy in this module.

Primary sources

1. Fahy E, Subramaniam S. [RefMet: a reference nomenclature for metabolomics](#). *Nature Methods*. 2020;17:1173-1174.
2. Metabolomics Workbench. [REST Service](#), [RefMet](#), and [mwTab documentation](#). Accessed August 10, 2026.

3. Wilkinson MD, Dumontier M, Aalbersberg IJ, et al. [The FAIR Guiding Principles for scientific data management and stewardship](#). *Scientific Data*. 2016;3:160018.
4. Sumner LW, Amberg A, Barrett D, et al. [Proposed minimum reporting standards for chemical analysis](#). *Metabolomics*. 2007;3:211-221.

Appendix - PCA worked examples

These figures are the exact worked examples generated by notebook Lesson 4. Find the corresponding code at NB-L4-PCA-DIET and NB-L4-PCA-EXERCISE in module/notebooks/metabo_diet_harmonization.ipynb. Each model has its own coordinate system; do not compare point positions or axis values across figures.

Figure A1 - Diet-study plasma PCA

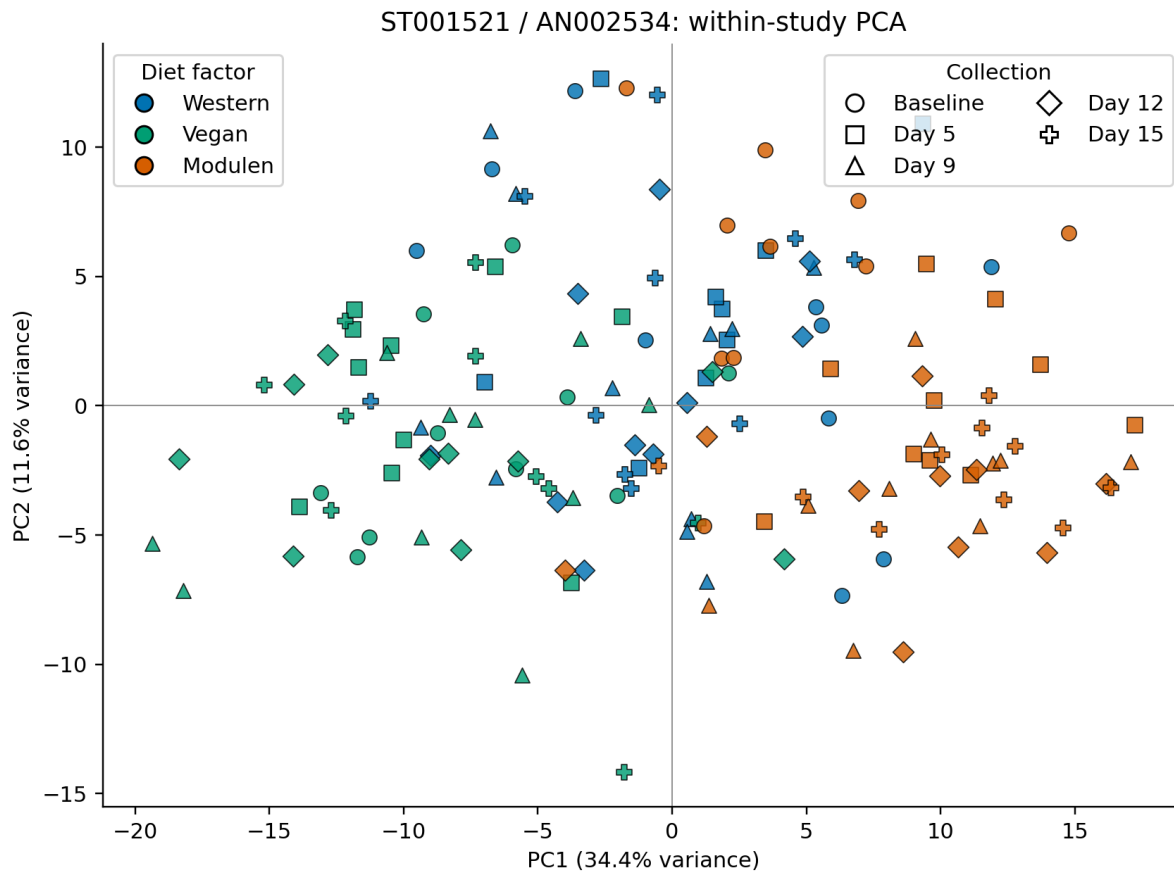


Figure A1. ST001521 / AN002534 within-study PCA of 150 biological plasma samples. The 10 pooled QPP samples are excluded.

How to read it: describe clustering or overlap as exploratory structure. The plot does not isolate a diet effect because setting, prior diet, antibiotics, PEG, time, and other design features can contribute.

Figure A2 - Exercise-study serum PCA

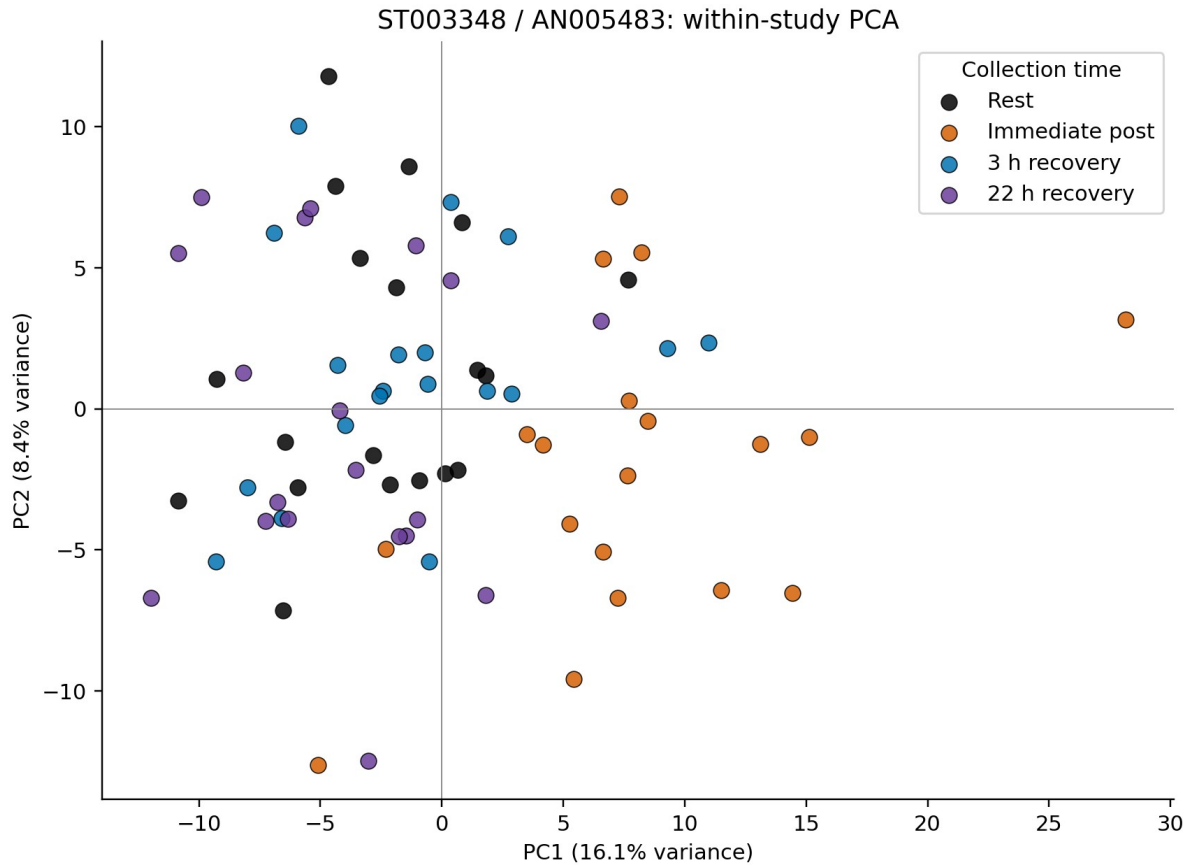


Figure A2. ST003348 / AN005483 within-study PCA of 76 serum samples after excluding explicit isotope-labeled/internal-standard rows.

How to read it: the timepoint pattern is exploratory and can also reflect fasting, clock time, repeated measures, or analytical structure. PCA alone is not a repeated-measures test and does not prove causation.

Appendix - Data provenance

This appendix summarizes the versioned evidence trail used by the release. The machine-readable source is `module/data/provenance.json`.

Provenance field	Release value
Repository	Metabolomics Workbench / National Metabolomics Data Repository
Retrieval time (UTC)	2026-08-10T11:44:11Z
Access model	open; no login, registration, credential, or data-use agreement required for the selected public REST records
Selection status	locked
Manifest schema	1.0.0

Locked study pair

Role	Accession	Specimen	Design summary	License
Diet	ST001521	Blood (plasma)	FARMM longitudinal diet study. Twenty omnivores were randomized to inpatient Western/omnivore or exclusive enteral nutrition (Modulen) feeding; ten established vegans continued their usual diet as outpatients. Plasma was collected at baseline and days 5, 9, 12, and 15.	CC BY 4.0
Exercise	ST003348	Blood (serum)	Acute endurance race-walking study with fasting pre-exercise REST, immediate post-exercise STAT, 3-hour recovery REC3, and fasting 22-hour recovery REC22 serum samples.	CC BY 4.0

RefMet overlap audit

Measure	Value
Diet unique nonblank RefMet names	510
Exercise unique nonblank RefMet names	475
Raw exact overlap	153
Known isotope/internal-standard exclusions	8
Conservative biological overlap	145

The 145-name count excludes eight known labeled/internal-standard mappings and is the operational biological crosswalk count. Further annotation curation may reduce it. Overlap supports identifier-

crosswalk and class-summary exercises; it does not make raw peak areas quantitatively comparable across studies.

Sanitization and claim boundaries

Files in module/data/raw are exact source snapshots and must not be edited in place. Sanitized/derived tables must retain source IDs and record these transformations.

St001521 Factors

- Trim factor-field whitespace without changing repository labels.
- Exclude QPP01 through QPP10 and rows whose Study_Diet, Sex, and Time factors are NA from biological analyses; retain them only as pooled-QC records.
- Derive biological participant_id from the leading four digits of local_sample_id and verify observed visits rather than forcing a balanced panel.
- Preserve Western as the factor label and document that the narrative calls this arm omnivore.
- Use the valid split study REST endpoints; do not ingest malformed AN002533, AN002535, or AN002536 mwTab JSON.

St003348 Factors

- Derive participant_id from the integer before the underscore in local_sample_id.
- Validate suffixes 1, 2, 3, and 4 against rest, stat, rec3, and rec22, respectively.
- Retain source sample_source=blood while setting harmonized specimen to serum from the collection block.
- Treat the treatment-block value no treatment as a metadata defect; define the exercise contrast from the documented protocol, collection, and time factors.

Cross Study Metabolites

- Remove the eight explicit isotope-labeled/internal-standard mappings listed in primary_pair.overlap before constructing the biological RefMet crosswalk.
- Keep analysis_id attached to duplicated RefMet names unless a documented within-study collapse rule is applied.
- Never concatenate uncalibrated peak areas across the plasma and serum studies.

Cached-source integrity

Study	Cached file	Bytes	SHA-256
ST001521	ST001521_summary.json	575	d2199dc631f33c09b9d9cb13f64a9c5233e1b6b7fe3f372e4cb8e947817541d9

Study	Cached file	Bytes	SHA-256
ST001521	ST001521_factors.json	49194	7b88427c7434f20538aff398f8ddac7e88d666696f4e19992c85d6017fe0372f
ST001521	ST001521_analysis.json	2025	72d60232e8a0f39eb24c790cabe7bd1585a1e852eb58f4b0288703273a0c70ac
ST001521	ST001521_metabolites.json	95806	7686f09d81f632d3996d5a5eceaedf31632a3c160d291f21cfa574d680251a53
ST001521	ST001521_data.json	1997104	5528f4a721189f45de65d65b325727177bc5355181f571827f6030a7e7676c06
ST003348	ST003348_summary.json	621	3766073c903b8975c2959fbb9c562b4cee531cddc02e749f380915354b193bee
ST003348	ST003348_factors.json	11791	e31bb851873fee1d185a2d91d8b5553637145d799793d416ecf2be6c3f8c3329
ST003348	ST003348_analysis.json	1051	27f654c3cee086b2310ebe4aeee64f5a835949e99c8b0da4736e6316652c4a19
ST003348	ST003348_metabolites.json	106651	492be3dd45c6bd741b64634249e4acc89a3d0c9750f98f816215305cff5ef566
ST003348	ST003348_data.json	1076605	0610a28e0e858524adefee5ee2d0dca46cf75c9a5bab1ebcb659e2573715adfb
RefMet	refmet_classification.json	31375639	b11b6f09c5f068226ac440b815c693af0255e506a2cb33736a265004a95cb50d

Data attribution and reuse

Primary public datasets

This training package includes cached copies and derived educational artifacts from two public Metabolomics Workbench/NMDR studies. Each study's public summary endpoint reports a Creative Commons Attribution 4.0 International license.

Diet-anchored case study

- Metabolomics Workbench study: [ST001521](#)
- Project: [PR001024](#)
- Project DOI: [10.21228/M8B984](#)
- Repository record: [Metabolomics Workbench ST001521](#)
- License: [CC BY 4.0](#)

Exercise-anchored case study

- Metabolomics Workbench study: [ST003348](#)
- Project: [PR002083](#)
- Project DOI: [10.21228/M8C802](#)
- Repository record: [Metabolomics Workbench ST003348](#)
- License: [CC BY 4.0](#)

RefMet classification

- Source: Metabolomics Workbench RefMet REST service
- Endpoint: [RefMet classification REST endpoint](#)
- Documentation: [Metabolomics Workbench REST documentation](#)

Exact retrieval times, response sizes, SHA-256 checksums, release fields, and transformations are recorded in [module/data/provenance.json](#) in the repository and analysis bundle.

Training materials

Except for third-party material identified here or in [module/data/provenance.json](#), the lesson text, original templates, notebook narrative and code, scripts, figures, and assessments in this package are licensed under the [Creative Commons Attribution 4.0 International Public License](#). Reusers must give appropriate credit, link to the license, and indicate changes. The repository and release packages include the full English legal code; it is also available from [Creative Commons](#).

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Interpretation notice

The package uses these studies to teach metadata reasoning, identifier harmonization, and within-study exploratory analysis. It does not treat the studies as one experiment, estimate a diet-versus-exercise causal effect, or claim that plasma and serum peak areas are quantitatively interchangeable.